

Analog revision sheet: Lectures 1–8

EEE/INSTR F313 mid-sem · every concept, formula and figure from your notes, in lecture order

0 · The toolkit: seven ideas that solve every question

One set of example numbers is used everywhere: $V_{DD} = 1.8\text{ V}$, $|V_{th}| = 0.5\text{ V}$, $|V_{ov}| = 0.2\text{ V}$, so $|V_{GS}| = 0.7\text{ V}$. (Lec 3–4 use Ex 9.7's 3 V supply.)

- **T1 · Saturation fences.** NMOS: $V_D \geq V_G - V_{th}$. PMOS: $V_D \leq V_G + |V_{th}|$. Equivalent channel form: $|V_{DS}| \geq |V_{ov}|$.
- **T2 · Check vs link.** A **check** is source→drain through a channel: it must be $\geq |V_{ov}|$, and it is what fails. A **link** is source→gate: always exactly $|V_{GS}| = |V_{th}| + |V_{ov}|$; it only tells you where a gate sits. A fence = one check + one link written in one line.
- **T3 · Towers.** One block = one transistor; height = the voltage across it; blocks add up to V_{DD} ; no block shorter than its V_{ov} ; moving an input or output slides one boundary (one block grows, its neighbour shrinks); the limit is the moment a block is squeezed to its minimum (red). A “room” block is not a transistor, just spare voltage.
- **T4 · Gain by two looks.** $A_v = G_m(R_{up} \parallel R_{down})$, with $G_m \approx g_m$ of the input device.
- **T5 · Look-in resistances.** Into a drain (source grounded): r_O . Into a drain with R_S under the source: $\approx g_m r_O R_S$ (**up multiplies**). Into a source with R_D on the drain: $\frac{R_D + r_O}{1 + g_m r_O} \approx \frac{1}{g_m}$ (**down divides**). Into a gate: ∞ .
- **T6 · Current divider.** A current reaching a node splits inversely to resistance: $I_{R_2} = I \frac{R_1}{R_1 + R_2}$. Signal current goes into a source ($1/g_m$), not into an r_O .
- **T7 · CM = the average of a pair.** $V_{in,CM} = \frac{V_{in1} + V_{in2}}{2}$, $v_d = V_{in1} - V_{in2}$. Output CM = average of the two outputs. A pole sits at every high-resistance node: $\omega_p = \frac{1}{RC}$.

Starting at Lec 6? The four tools every step uses

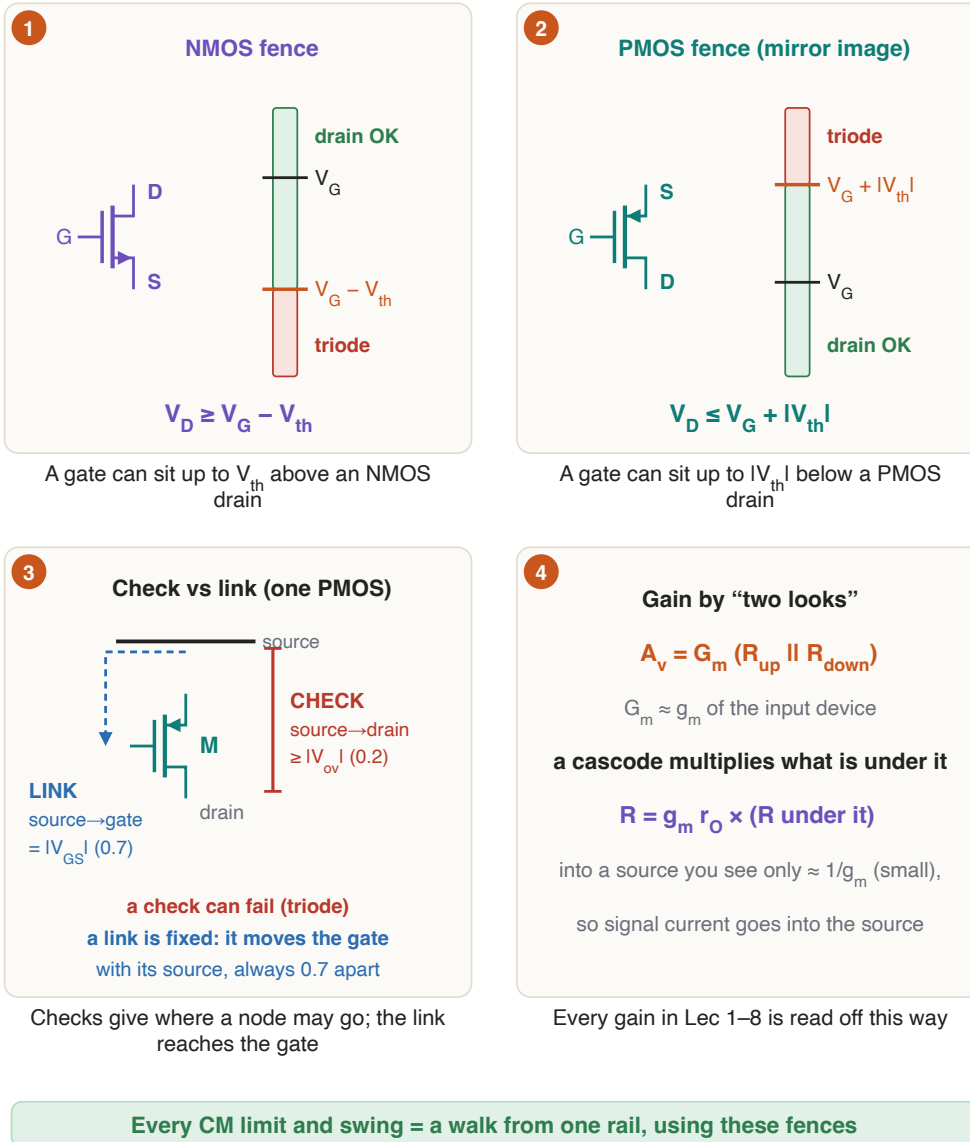


Fig 1. Starting at Lec 6: the two fences, check vs link, and gain by two looks.

How to read a tower (Ex 9.7 numbers: $V_{DD} = 3\text{ V}$)

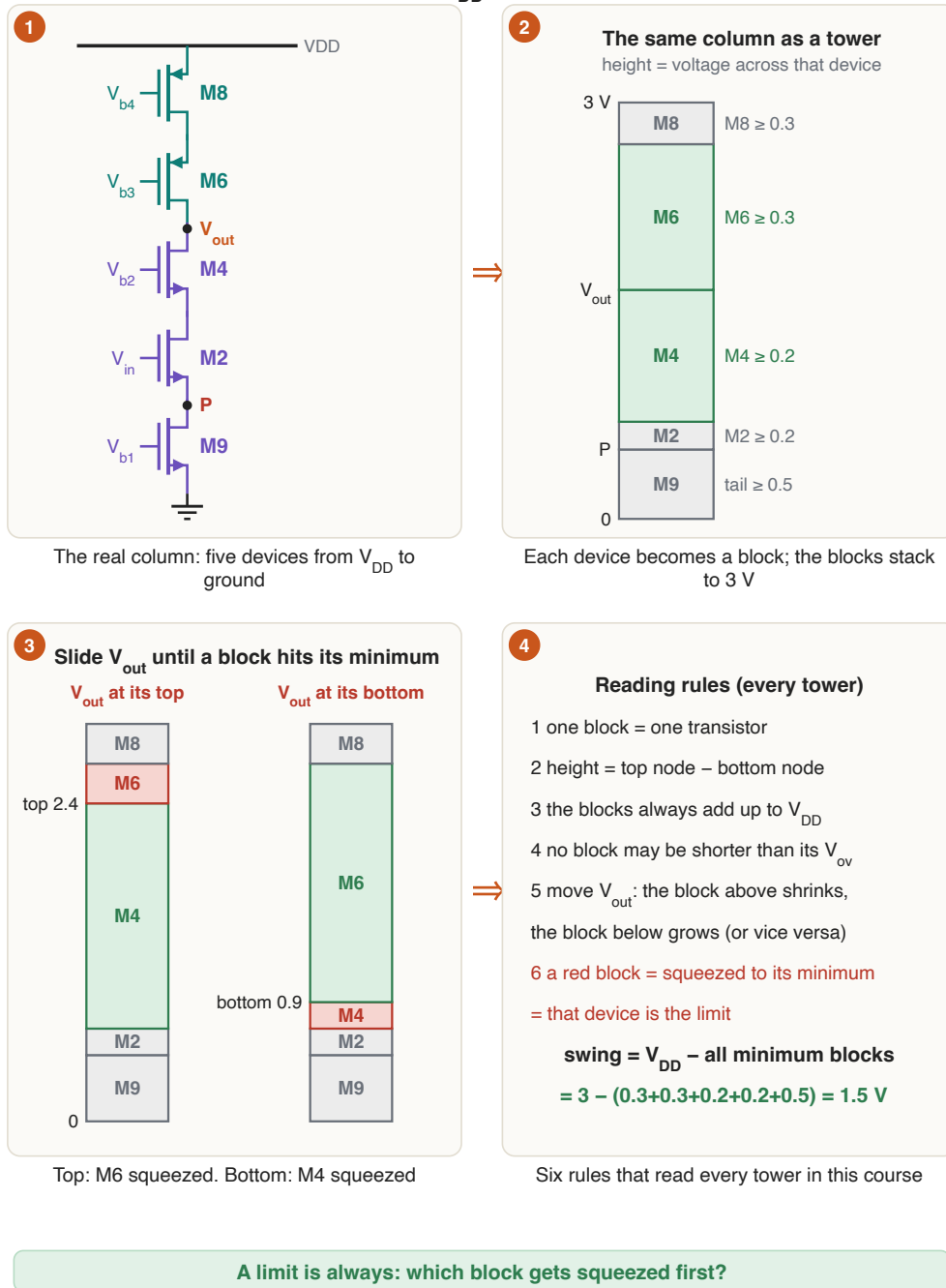
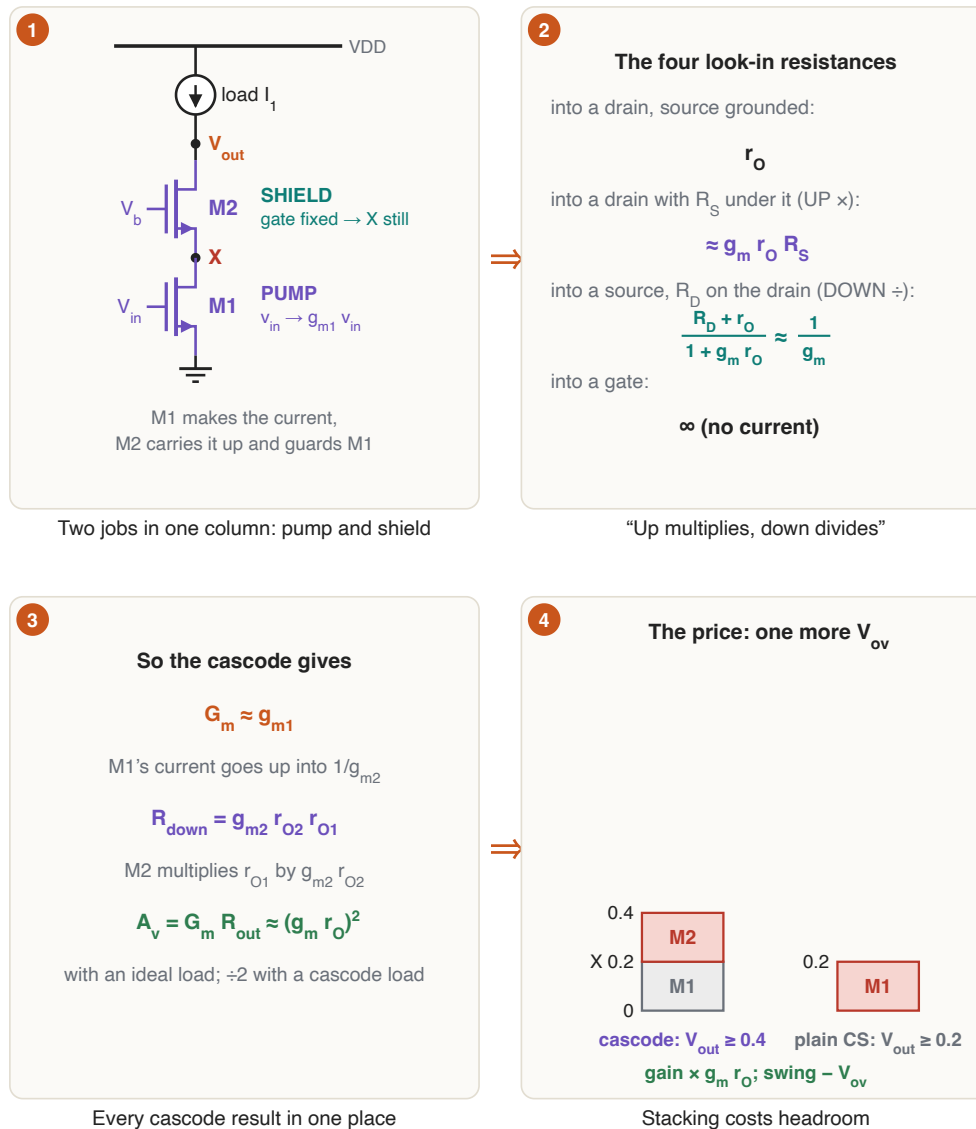


Fig 2. How to read a tower: one block per transistor, height = voltage across it, a red block is the squeezed device that sets the limit.

Cascode refresher (everything Lec 5–8 builds on)



Gain = $G_m \times R_{\text{out}}$: the cascode keeps G_m and multiplies R_{out}

Fig 3. Cascode refresher: pump and shield, the four look-in resistances, G_m , R_{out} and the headroom price.

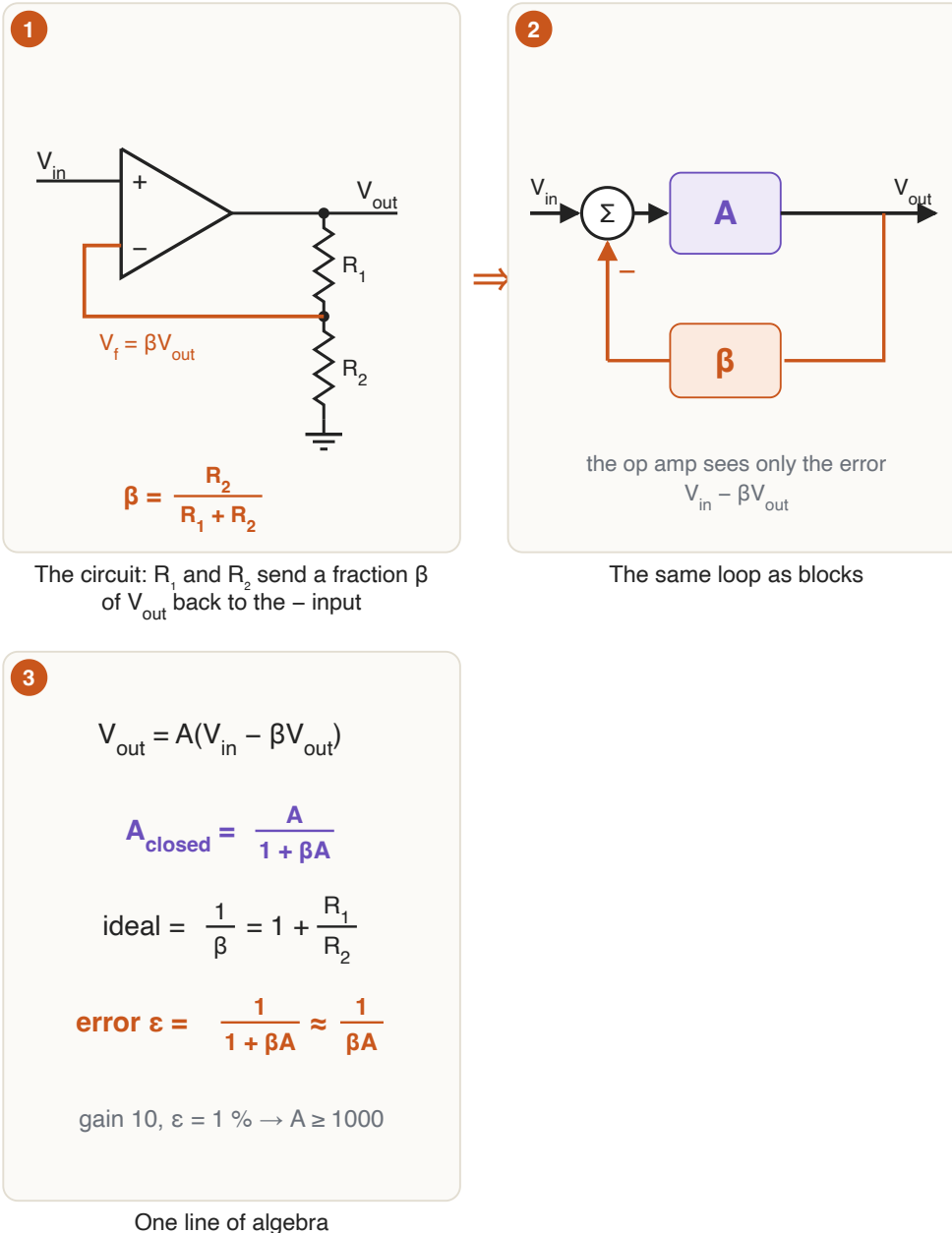
Exam recipes

1. **CM range / swing:** move the input (or output) towards one rail $\rightarrow$ find the block that gets squeezed $\rightarrow$ its **check** gives the limit on a node $\rightarrow$ use the **link** to reach the gate.
2. **Gain:** find G_m (the input device's g_m ; check the current divider) $\rightarrow$ look up $\rightarrow$ look down $\rightarrow A_v = G_m(R_{\text{up}} \parallel R_{\text{down}})$.
3. **Bandwidth:** find the high-resistance node, $\omega_p = 1/(R_{\text{out}} C_L)$; Hz = divide by 2π at the end.

Lec 1 · Gain and gain error (Razavi Ex 9.1)

A one-transistor gain $-g_m R_D$ drifts with process and temperature. Feedback fixes it: a huge, sloppy open-loop gain A plus two resistors, whose **ratio** is precise. The resistors set the gain; A only decides how close you get.

Feedback in three pictures



Resistors set the gain ($1/\beta$); A only decides how close you get ($\varepsilon \approx 1/\beta A$)

Fig 4. Feedback in three pictures: the circuit, the same loop as blocks, and the one line of algebra.

feedback factor (tap across the bottom resistor)

$$\beta = \frac{R_2}{R_1 + R_2}$$

closed-loop gain

$$A_{closed} = \frac{A}{1 + \beta A}$$

ideal gain

$$A_{ideal} = \frac{1}{\beta} = 1 + \frac{R_1}{R_2}$$

gain error

$$\varepsilon = \frac{1}{1 + \beta A} \approx \frac{1}{\beta A}$$

minimum open-loop gain

$$A_{min} = \frac{A_{closed}}{\varepsilon}$$

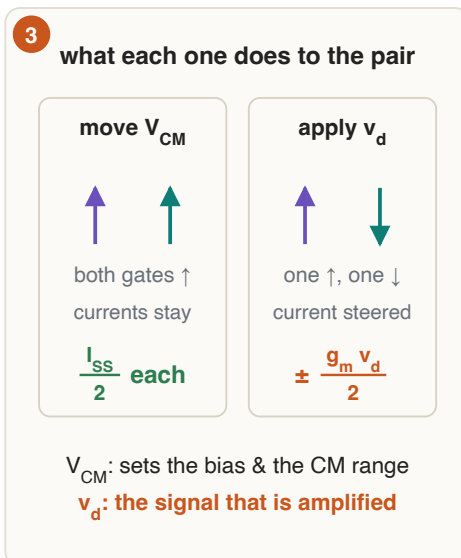
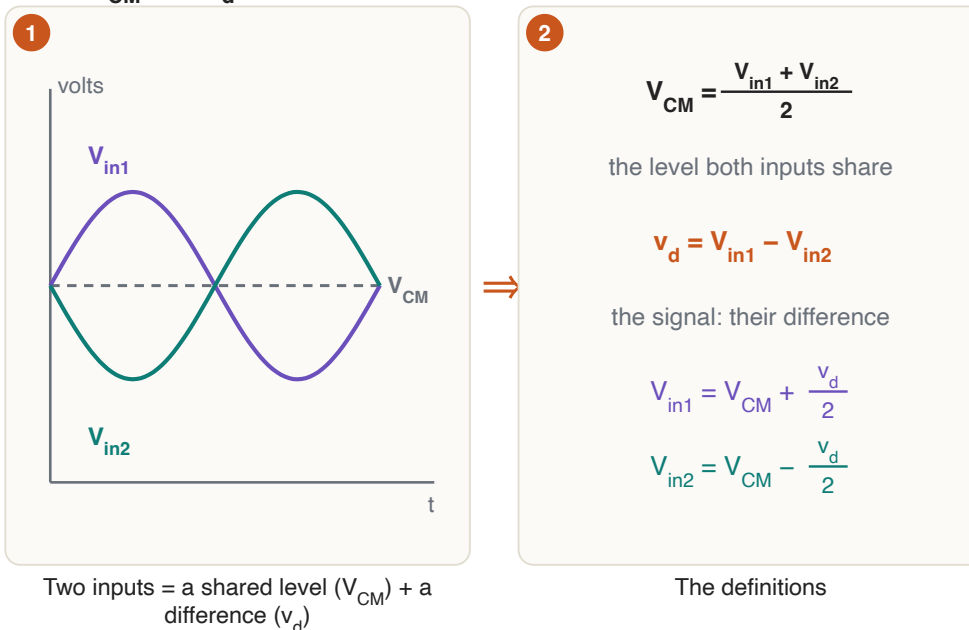
Watch out

- Gain 10 with 1 % error: $A \geq 1000$ (exact form gives 990; quote 1000 and say which).
- β uses the **bottom** resistor R_2 . $\text{dB} = 20 \log_{10}$.
- A buffer has $\beta = 1$, so its error is $\approx 1/A$.

Lec 2 · Specs, GBW, the differential pair and the 5-T OTA

Computed specs: **gain**, **bandwidth (GBW)**, **output swing**, **input CM range**. One pole: gain $\times$ bandwidth is constant. Current-source loads give a large r_O for only $|V_{ov}|$ of DC room.

What V_{CM} and v_d mean

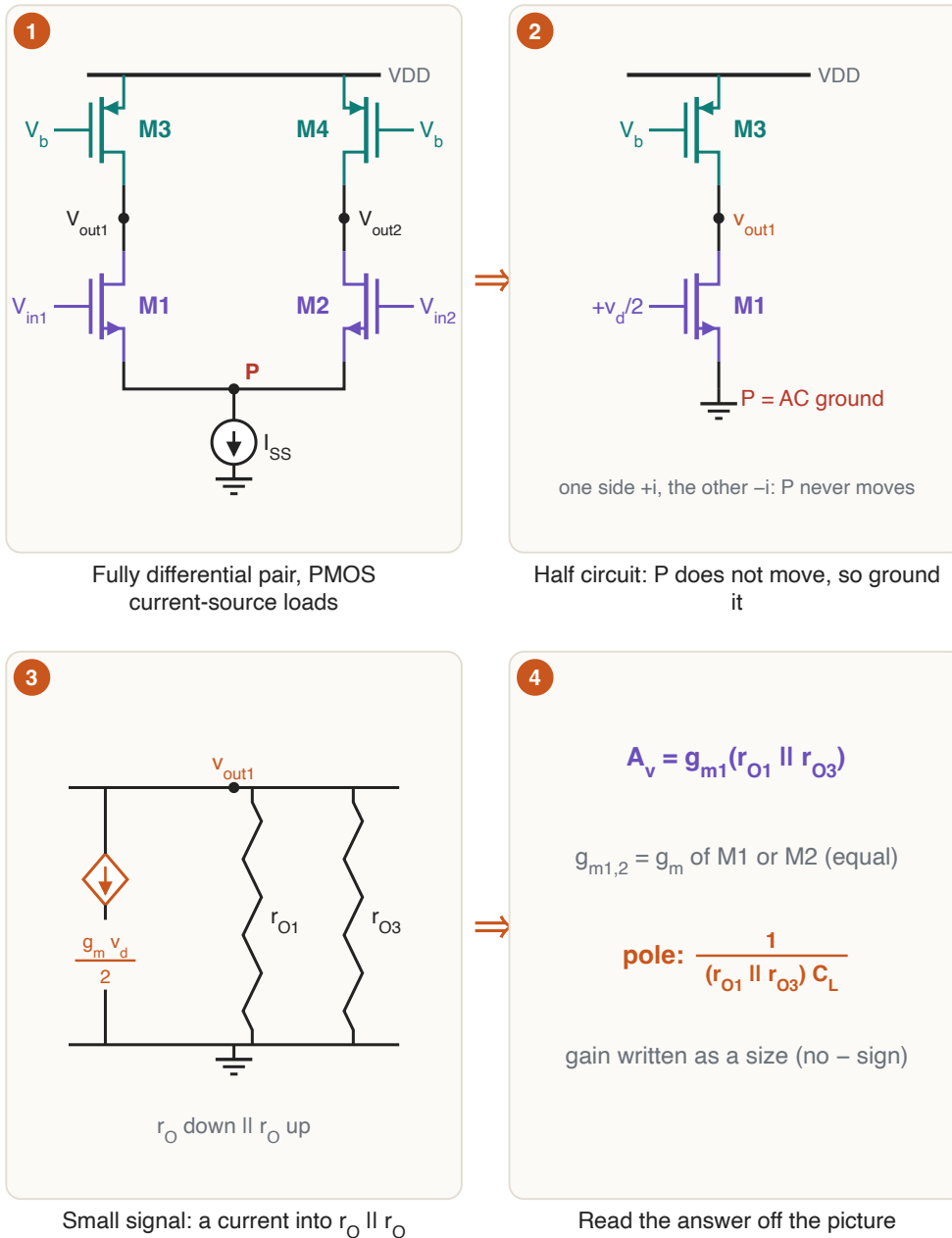


The tail fixes the sum, so only v_d steers current

Common mode = the average of the two inputs; differential = their difference

Fig 5. What V_{CM} and v_d mean: the shared level and the difference, and what each does to the pair.

Differential pair: full circuit $\Rightarrow$ one line



Every gain: $A_v = G_m \times R_{out}$ (current made $\times$ resistance it meets)

Fig 6. Differential pair simplified: full circuit $\Rightarrow$ half circuit (P is AC ground) $\Rightarrow$ small-signal picture $\Rightarrow A_v = g_m(r_{O1} \parallel r_{O3})$.

gain-bandwidth product

$$\omega_u = A_0 \omega_0 = \text{GBW}$$

its input CM range

$$V_{ISS} + V_{GS1} \leq V_{in,CM} \leq V_{DD} - |V_{ov3}| + V_{thn}$$

output pole

$$\omega_p = \frac{1}{(r_{O2} \parallel r_{O4})C_L}$$

fully differential pair, PMOS current-source loads

$$A_v = g_{m1,2}(r_{O1,2} \parallel r_{O3,4})$$

its differential swing ($\times 2$: outputs move oppositely)

$$2(V_{DD} - |V_{ov3}| - V_{ov1} - V_{ISS})$$

Why the differential swing is doubled

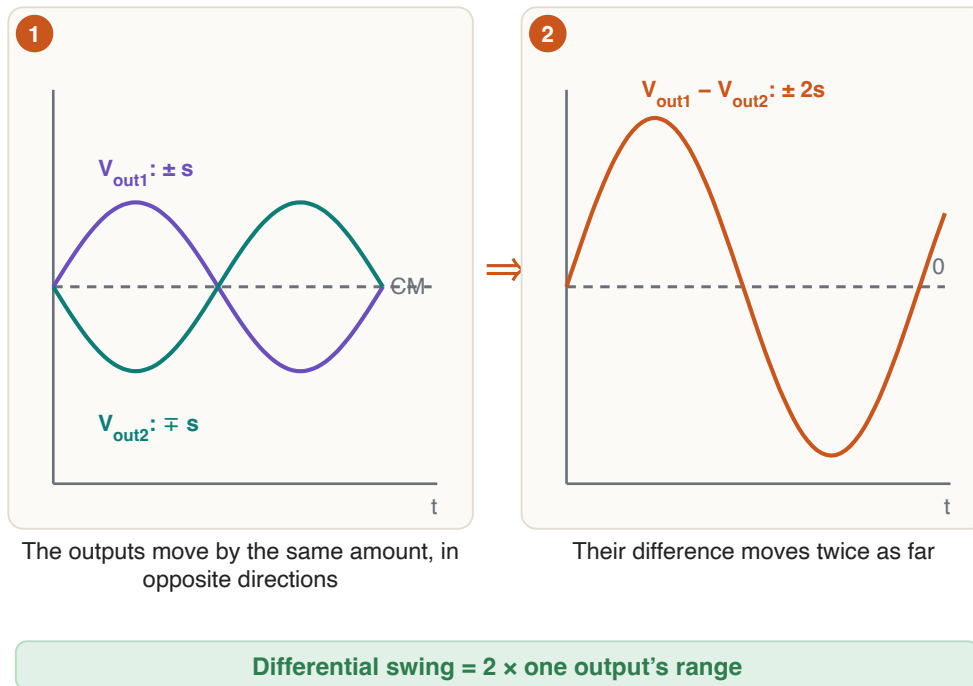
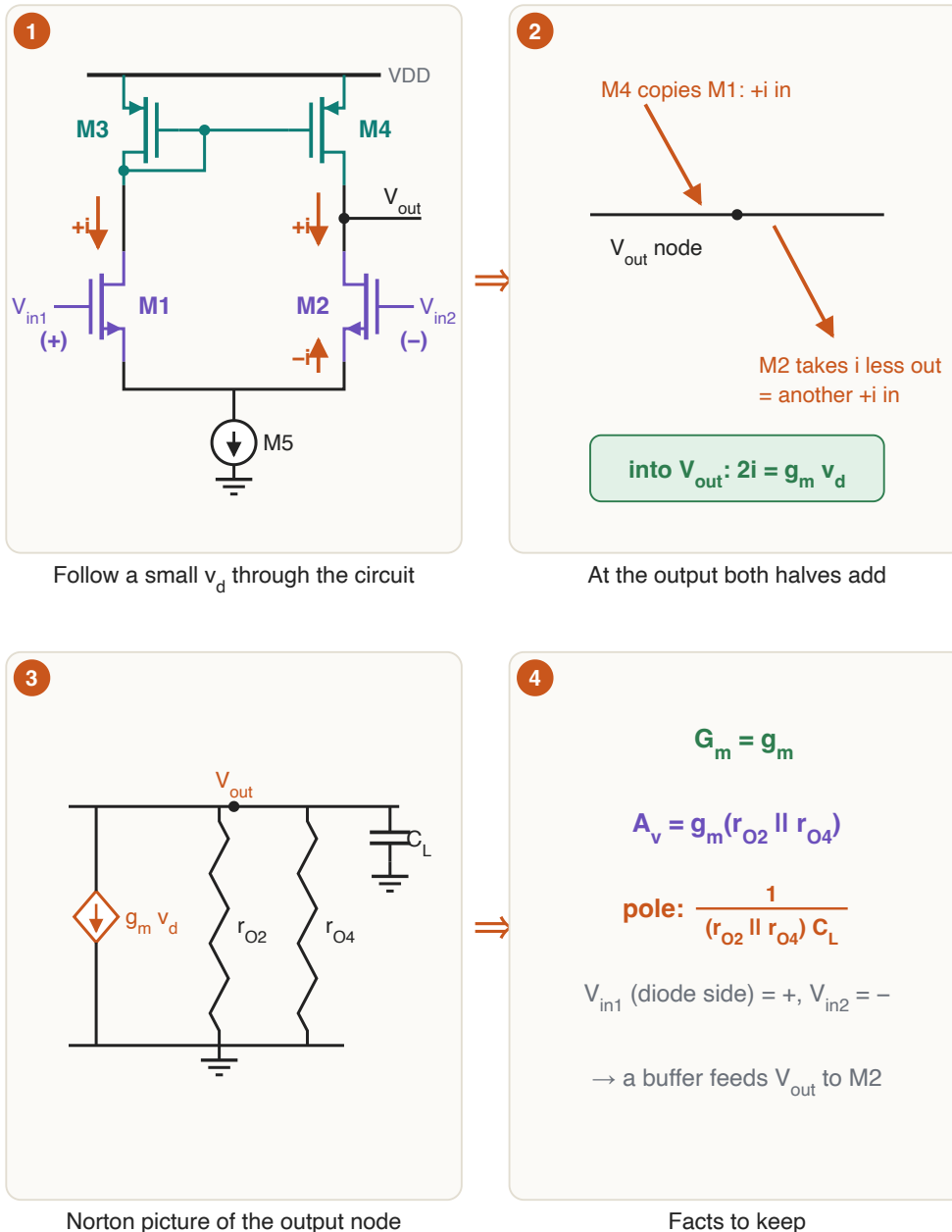


Fig 7. Why the differential swing is doubled: the two outputs move oppositely, so their difference moves twice as far.

5-T OTA: M3 is a diode, M4 mirrors it; both halves' currents add at the single output, so the full $g_m v_d$ arrives. **M2's gate is the inverting (–) input.**

Five-transistor OTA: where the signal current goes



The mirror adds the two halves: $A_v = g_m(r_{O2} \parallel r_{O4})$

Fig 8. 5-T OTA simplified: follow the signal current, add the halves at the output, then the Norton picture of the output node.

5-T OTA gain

$$A_v = g_{m2}(r_{O2} \parallel r_{O4})$$

CM ceiling (diode node, then one V_{th} up)

$$V_{in,CM,max} = V_{DD} - |V_{GS3}| + V_{th1}$$

CM floor (tail, then a whole V_{GS})

$$V_{in,CM,min} = V_{ISS} + V_{GS1}$$

single-ended swing (no $\times 2$)

$$V_{DD} - |V_{ov4}| - V_{ov2} - V_{ISS}$$

5-T OTA: CM range and swing as towers

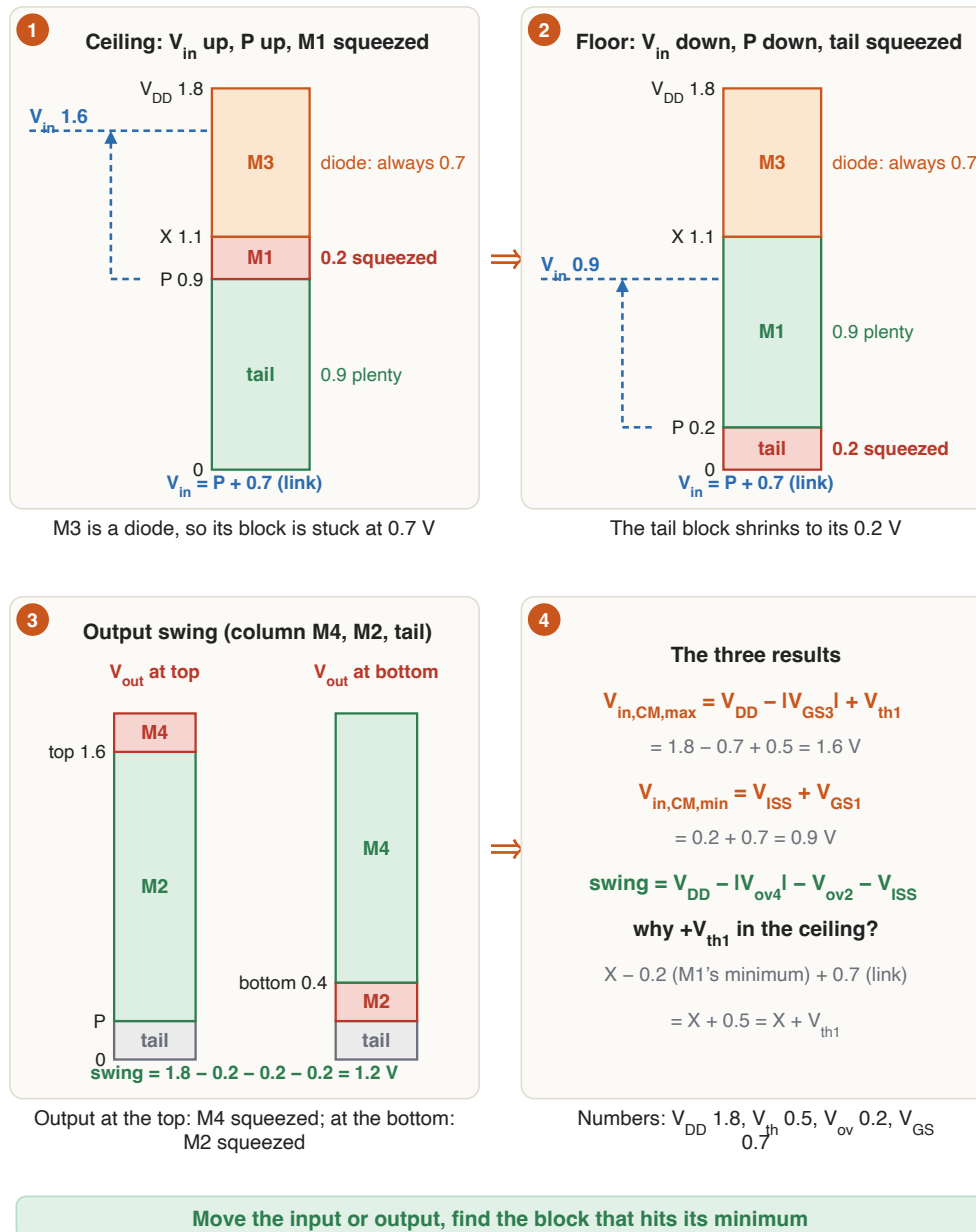


Fig 9. 5-T OTA CM range and swing as towers: move the input or output and find the block that gets squeezed.

Watch out

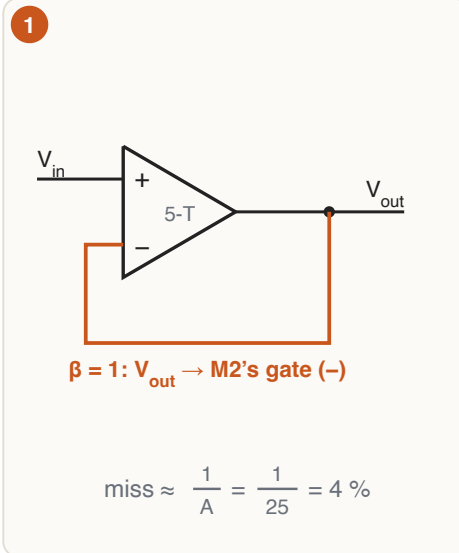
- A diode costs a whole $|V_{GS}|$; a current source only $|V_{ov}|$.
- $V_{out,min} = V_{ISS} + V_{ov2}$ assumes the input CM at its lowest; exactly it is $V_{in,CM} - V_{th2}$.
- rad/s vs Hz: divide by 2π at the end.

Asked in: Mid-sem Quiz 1 Part C: sizes from the CM limits, gain, swing, f-3dB with 4 pF.

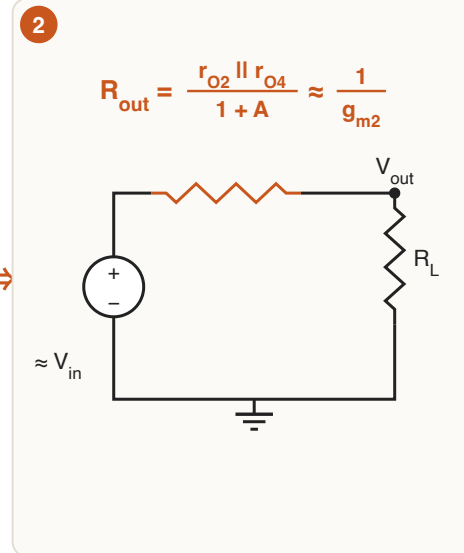
Lec 3 · Unity-gain buffer, telescopic cascode, the buffer window

Buffer: output wired to the $-$ input, $\beta = 1$, $V_{out} \approx V_{in}$. Feedback makes the output $(1 + \beta A)$ times stiffer and $(1 + \beta A)$ times faster.

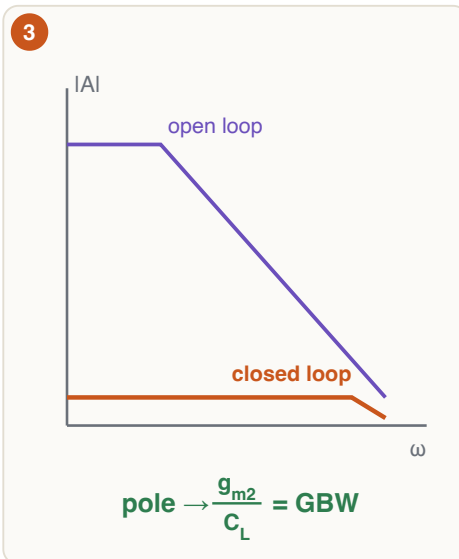
Unity-gain buffer: why $V_{out} = V_{in}$, stiff and fast



The 5-T OTA as a buffer: feedback to the - input (M2)



What a load sees: V_{in} behind a tiny resistance



Same C_L , R shrank by $(1 + A)$: faster

Feedback divides R_{out} by $(1 + \beta A)$ and multiplies the bandwidth by the same factor

Fig 10. Unity-gain buffer: feedback to M2, the load's view (V_{in} behind $\approx 1/g_m$), and the pole moving up to g_m/C_L .

buffer gain

$$A_{closed} = \frac{A}{1 + A} \approx 1$$

closed-loop output resistance

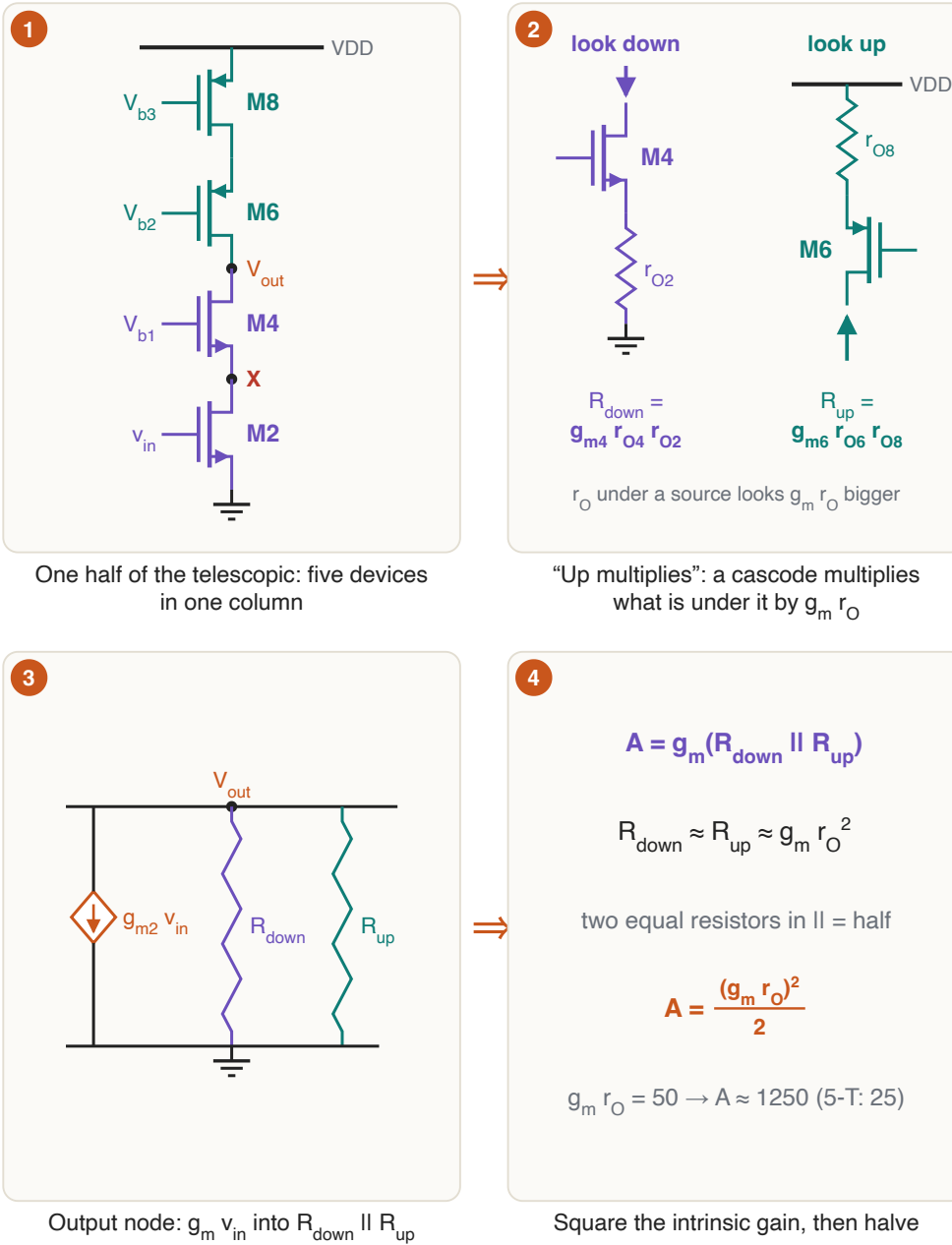
$$R_{out,closed} = \frac{R_{out,open}}{1 + \beta A} \approx \frac{1}{g_{m2}}$$

buffer bandwidth = the GBW

$$\omega = \frac{g_{m2}}{C_L}$$

Telescopic cascode: cascode both the pair and the loads. Look down: $g_{m4}r_{o4}r_{o2}$; look up: $g_{m6}r_{o6}r_{o8}$; two similar values in parallel halve.

Telescopic cascode gain, simplified



Cascode keeps $G_m = g_m$ and multiplies R_{out} by $g_m r_O$ on both sides

Fig 11. Telescopic gain simplified: one column $\Rightarrow$ look down / look up (“up multiplies”) $\Rightarrow g_m$ into $R_{down} \parallel R_{up} \Rightarrow (g_m r_O)^2/2$.

telescopic gain

$$A = g_{m1,2} [g_{m4} r_{O4} r_{O2} \parallel g_{m6} r_{O6} r_{O8}] \approx \frac{(g_m r_O)^2}{2}$$

differential swing (five blocks per column)

$$2 [V_{DD} - (|V_{ov8}| + |V_{ov6}| + V_{ov4} + V_{ov2} + V_{ISS})]$$

mirror-loaded version (diode tax)

$$V_{DD} - (|V_{ov8}| + |V_{ov6}| + |V_{thp}| + V_{ov4} + V_{ov2} + V_{ISS})$$

buffer window (V_{out} on M2's gate)

$$V_{b1} - V_{th4} \leq V_{out} \leq V_{b1} - V_{GS4} + V_{th2}$$

window width

$$V_{th} - V_{ov4}$$

Gain costs headroom

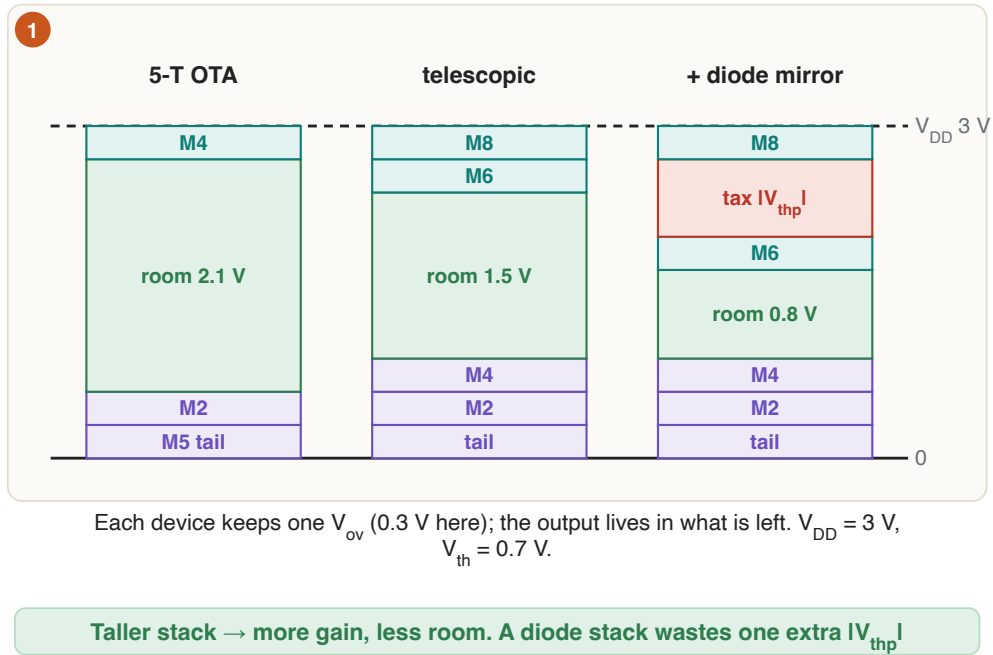
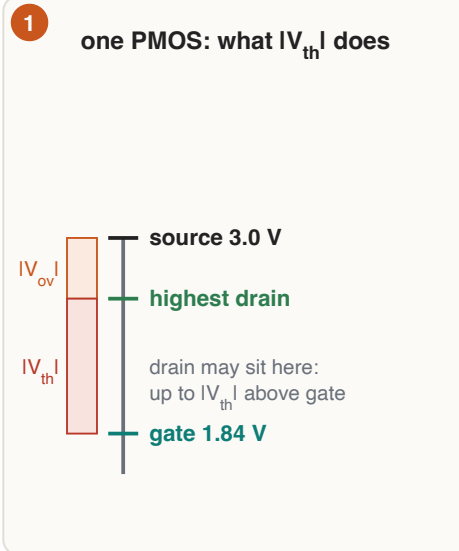
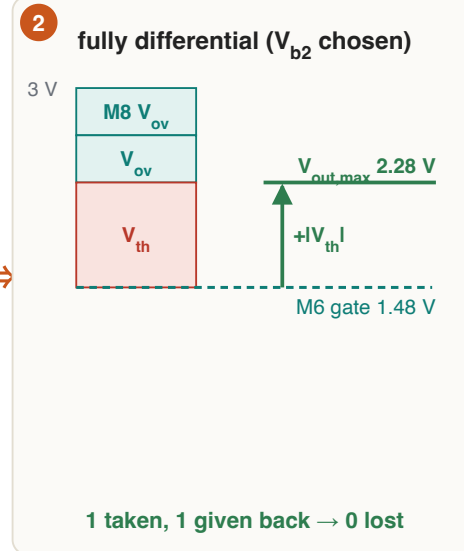


Fig 12. Gain costs headroom (towers): every stacked device keeps one V_{ov} block; a diode stack wastes one extra $|V_{thp}|$ (the diode tax).

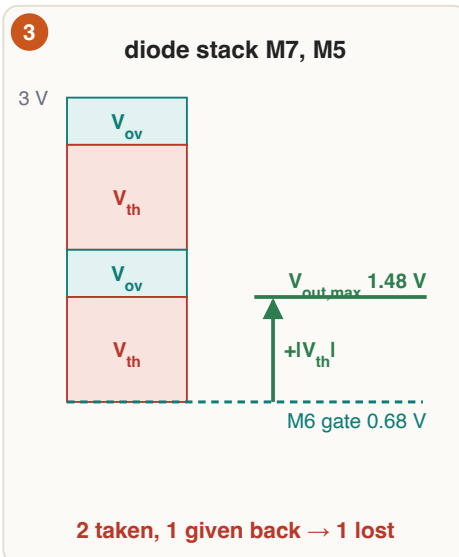
The diode tax: count the V_{th} 's



The gate pays $|V_{th}|$; the drain only needs $|V_{ov}|$



Walk down to M6's gate, then back up to its drain

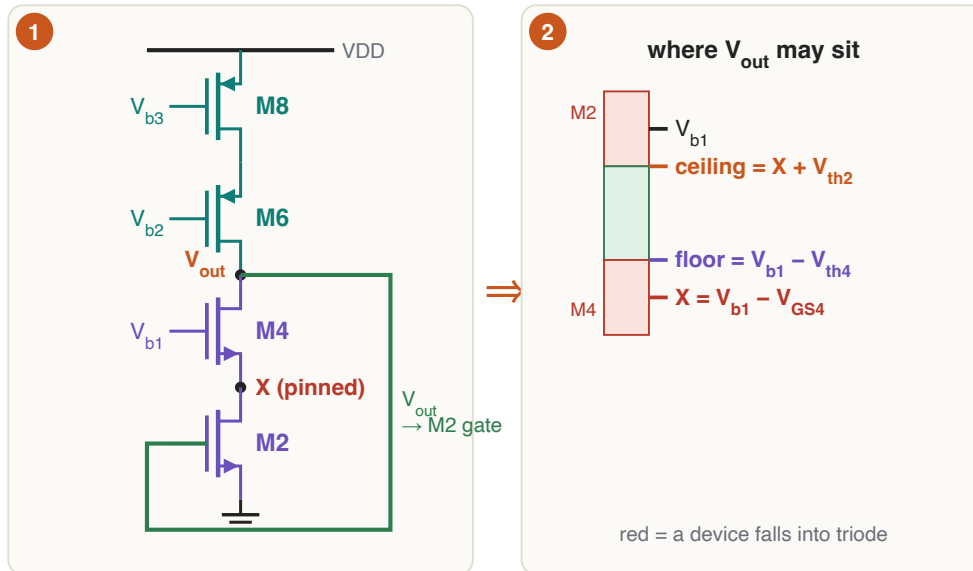


Two diodes: one $|V_{th}|$ never comes back
= the diode tax

Each V_{GS} on the way down costs a V_{th} ; the drain gives one back. Leftover = the tax

Fig 13. The diode tax: the gate pays $|V_{th}|$, the drain gives one back; with two diodes one $|V_{th}|$ is never recovered.

Telescopic as a buffer: a half-volt window



The telescopic as a buffer: V_{out} is also M2's gate

Two fences squeeze the output

3

$$\begin{aligned} \text{width} &= \text{ceiling} - \text{floor} \\ &= (V_{b1} - V_{GS4} + V_{th2}) - (V_{b1} - V_{th4}) \\ &= V_{th} - V_{ov4} \approx 0.5 \text{ V} \end{aligned}$$

V_{b1} cancels: it only slides the window

fix $\rightarrow$ fold the cascode (Lec 5)

The width of the window

M4 wants V_{out} high, M2 wants it low: only $V_{th} - V_{ov}$ is left

Fig 14. Telescopic as a buffer: M4 sets a floor, M2 a ceiling; the window is only $V_{th} - V_{ov}$ wide.

Watch out

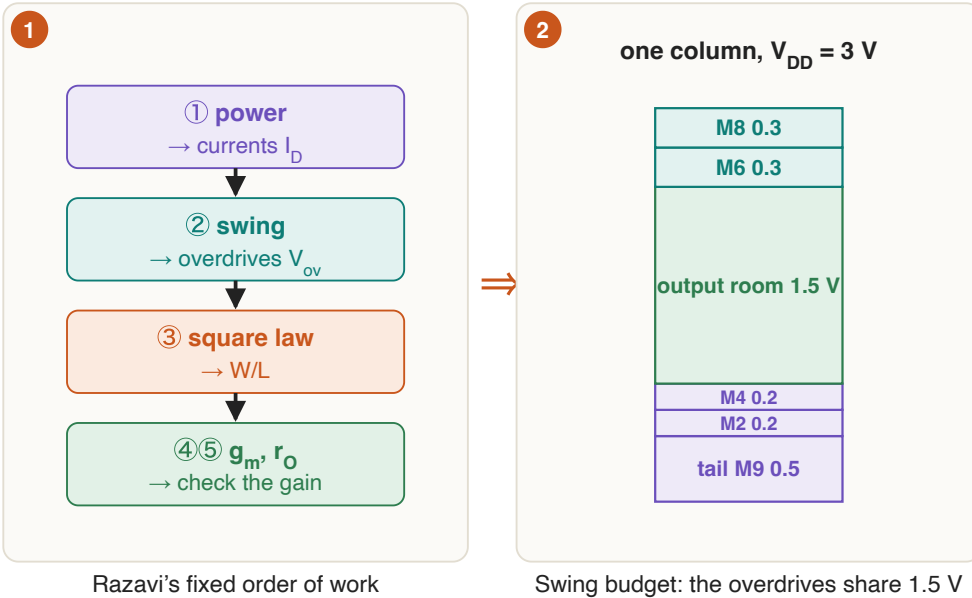
- In the window, M4 gives the **floor**, M2 the **ceiling**; V_{b1} only slides the window.
- X (M2's drain) = $V_{b1} - V_{GS4}$ is pinned; it is not the input CM.

Asked in: Mid-sem Q1(e) (buffer bandwidth $gm_2/2\pi CL$); Tutorial 2 Q1–Q2; Tutorial 2 Q2(b), Tutorial 3 Q1(c) (window).

Lec 4 · Designing a telescopic op amp (Razavi Ex 9.7)

Fixed order: ① power $\rightarrow$ currents, ② swing $\rightarrow$ overdrives, ③ square law $\rightarrow$ W/L, ④⑤ $g_m, r_O \rightarrow$ check the gain, then fix the weak side.

Designing a telescopic (Ex 9.7)



3

$$g_m = \frac{2I_D}{V_{ov}}, \quad r_O = \frac{1}{\lambda I_D}$$

$R_{up} \ 111\text{ k} \parallel R_{down} \ 666\text{ k}$

smaller one wins → $A \approx 1428$ (too low)

double L of PMOS: λ halves

$R_{up} = 444\text{ k} \rightarrow A \approx 4000$

Check, then fix the weak side

Power → currents; swing → overdrives; square law → W/L; then check the gain

Fig 15. Ex 9.7 design flow: power → currents, swing → overdrives, square law → W/L, then check and fix the gain.

currents $I_{total} = \frac{P}{V_{DD}}$	swing budget (per column) $ V_{ov8} + V_{ov6} + V_{ov4} + V_{ov2} + V_{ov9} = V_{DD} - \text{swing per side}$
square law $\frac{W}{L} = \frac{2I_D}{\mu C_{ox} V_{ov}^2}$	small-signal $g_m = \frac{2I_D}{V_{ov}}, \quad r_O = \frac{1}{\lambda I_D}$
gain check $A_v = g_{m1,2}(R_{up} \parallel R_{down})$	fix without touching swing $g_m r_O \propto \sqrt{\frac{WL}{I_D}}, \quad \lambda \propto \frac{1}{L}$

Watch out

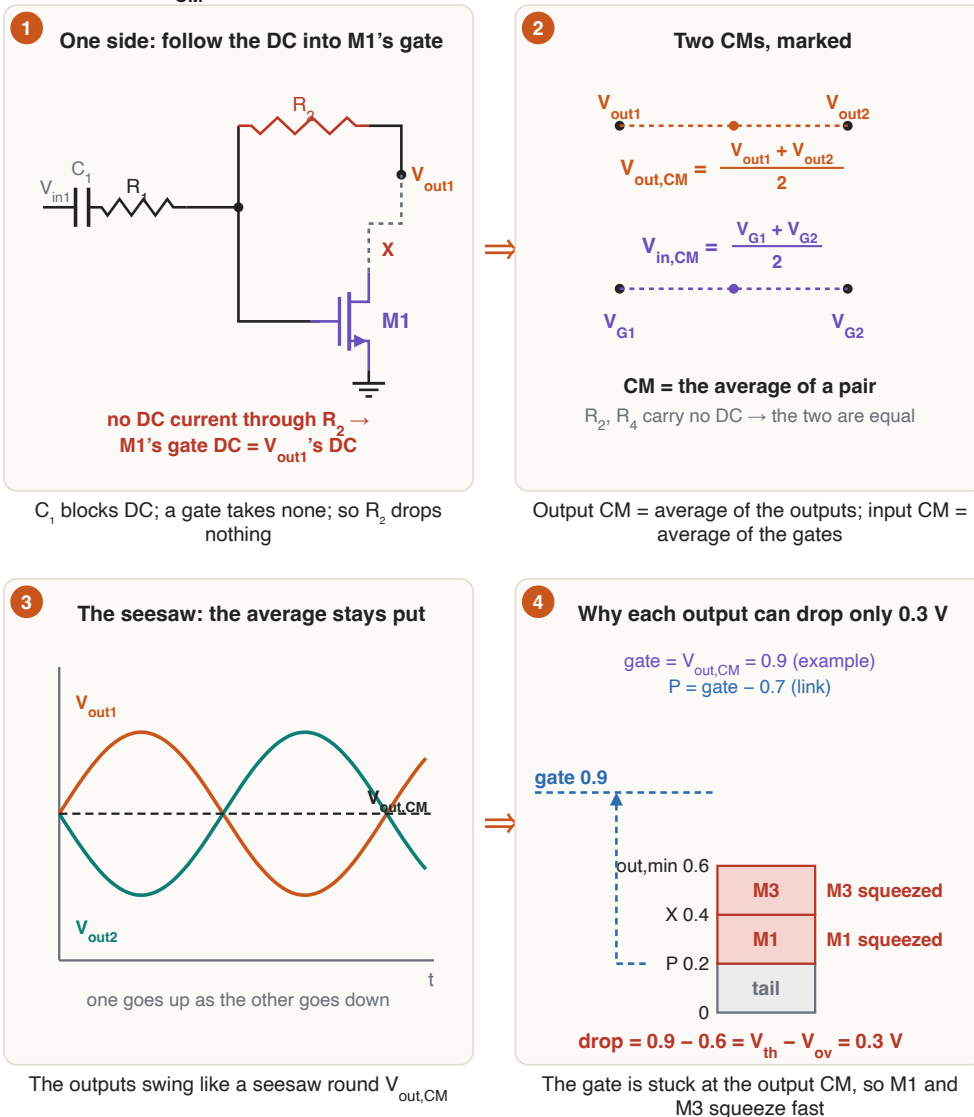
- Ex 9.7: 1.5 mA per side; V_{ov} : tail 0.5, PMOS 0.3, NMOS 0.2 V; gain ≈ 1428 (Razavi 1416) < 2000 .
- Double W **and** L of the PMOS only: W/L and V_{ov} stay, r_O doubles, gain ≈ 4000 .
- In parallel the smaller resistance wins (here R_{up}).

Asked in: Problem Set 1 P3, P6 and Tutorial 2 Q3: this recipe with new numbers.

Lec 5 · Ex 9.6, the folding transformation, the folded cascode

Ex 9.6: closed through capacitors, no DC flows in R_2 , so each gate sits at its output's DC level: **input CM = output CM**. M1 and M3 are then squeezed after a drop of only $V_{th} - V_{ov}$.

Ex 9.6: what V_{CM} means, and why the swing is squeezed

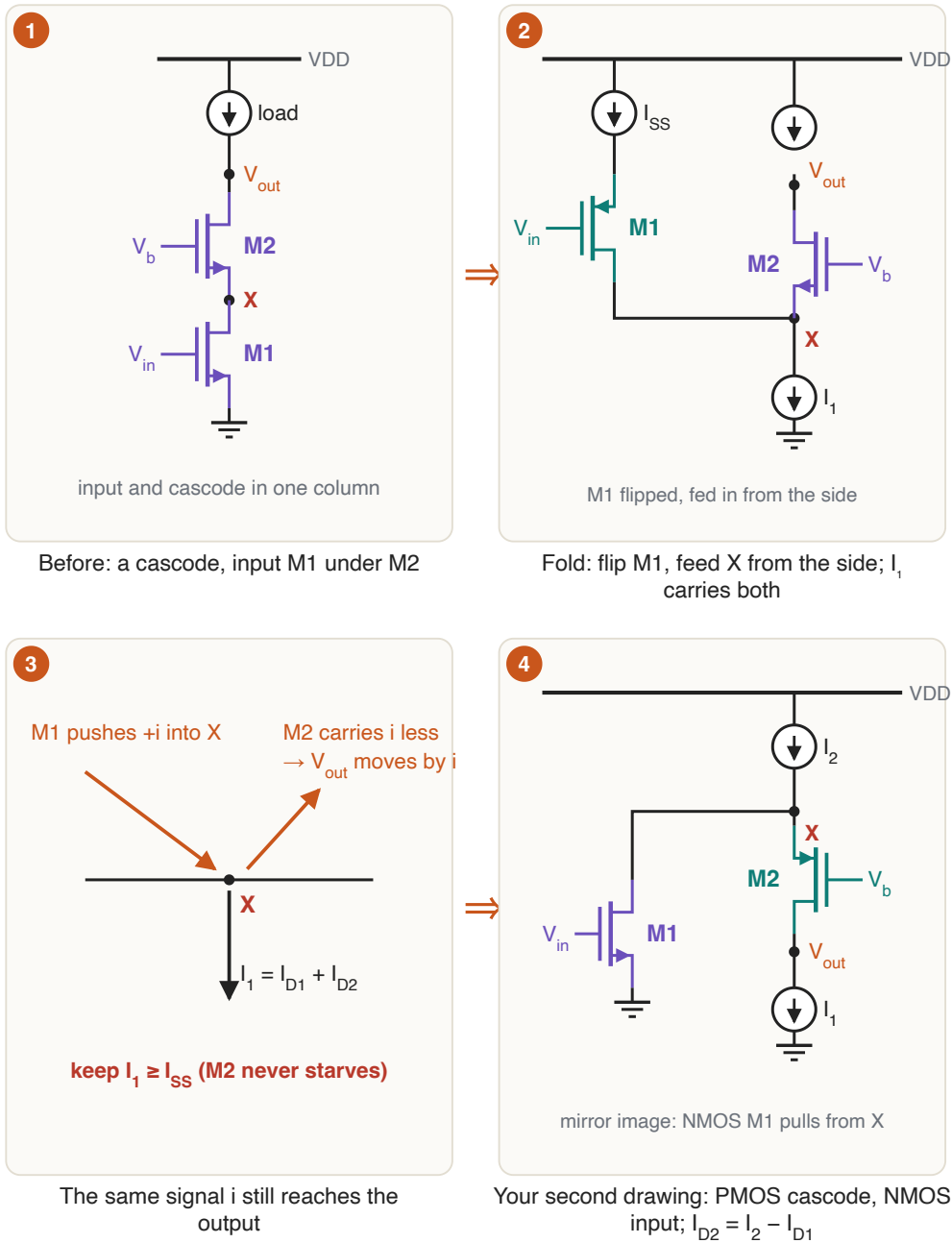


Input CM = output CM. Each output can fall only $V_{th} - V_{ov}$ below it

Fig 16. Ex 9.6: no DC through R_2 , both CMs marked, the output seesaw, and the tower showing why each output can drop only $V_{th} - V_{ov}$.

Folding: the cascode device only needs a current pushed into its source. Flip the input device and feed it in from the side, at the fold node; add a current source for the DC path. Same G_m , same formula, input out of the output column.

Folding: the input device leaves the output column



Same $G_m \approx g_{m1}$ — but the output column no longer holds the input pair

Fig 17. Folding: a cascode $\Rightarrow$ the input device flipped and fed from the side $\Rightarrow$ the same signal current reaches the output; panel 4 is the mirror version (NMOS input, PMOS cascode).

What folding changes, and what it does not

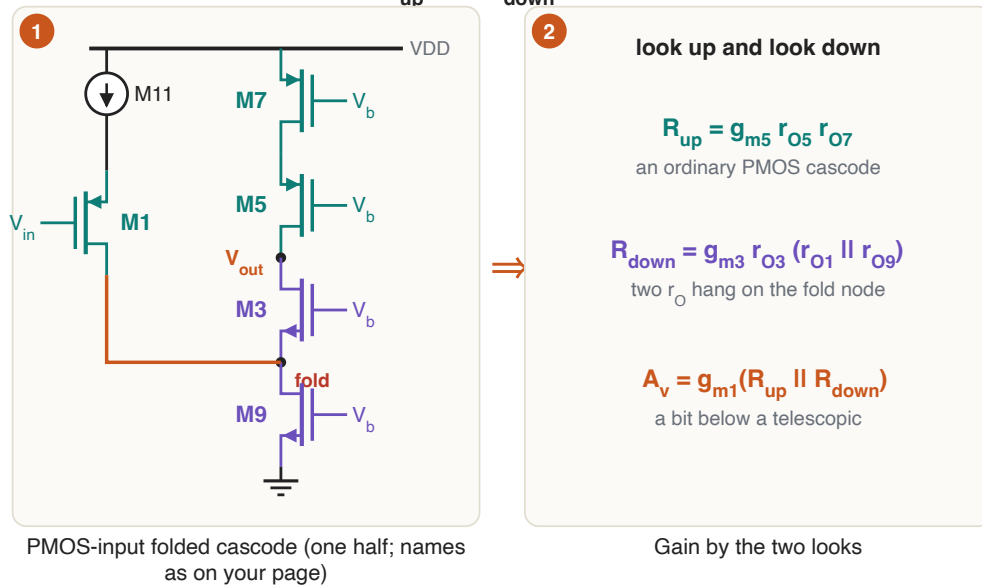
	normal cascode	folded cascode
1		
G_m	$\approx g_{m1}$	$\approx g_{m1}$ (same)
sign	inverting	inverting (same)
R_{down}	$g_{m2} r_{O2} r_{O1}$	$g_{m2} r_{O2} (r_{O1} \parallel r_{O,I2})$ (a bit lower)
X	$V_b - V_{GS2}$	$V_b - V_{GS2}$ (same)
input range	V_{GS1} to $X + V_{th1}$ (tight)	wide; reaches a rail
output column	load, M2, M1	load, M2, I_2 (input left it)
current	I_1	$I_1 + I_{D1} \approx \text{double}$ (costs power)
extra	—	pole at X, more noise
as a buffer	tiny window	works

Green = unchanged, red = what folding changes

M2 does not care where its current comes from: same G_m and same formula

Fig 18. Normal vs folded cascode: what folding changes and what it leaves alone.

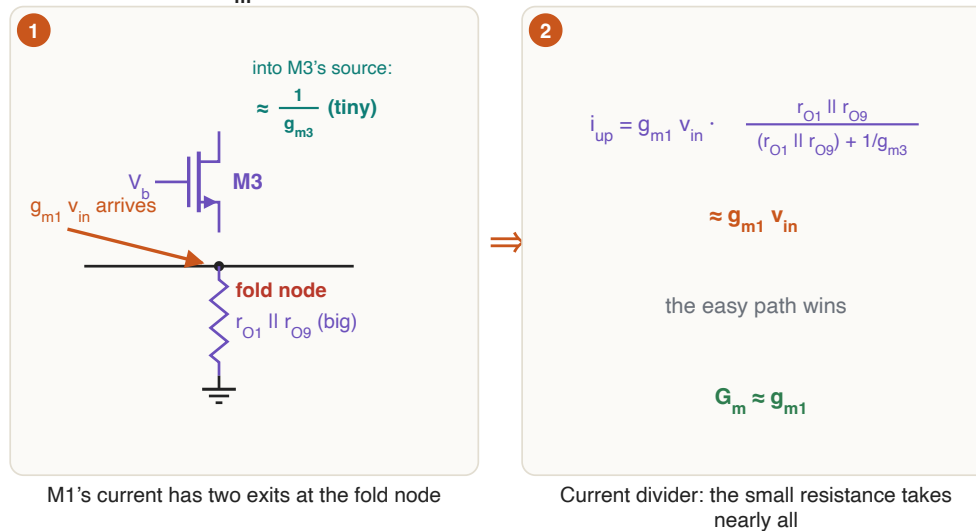
PMOS-input folded cascode, R_{up} and R_{down}



The fold node carries $r_{O1} \parallel r_{O9}$ — that is the only new thing

Fig 19. PMOS-input folded cascode: R_{up} is an ordinary cascode; R_{down} has $r_{O1} \parallel r_{O9}$ at the fold node.

Folded cascode G_m by inspection



Current takes the easy path: into a source ($1/g_m$), not into r_o

Fig 20. Folded cascode by inspection: the current divider at the fold node gives $G_m \approx g_{m1}$, then R_{up} , R_{down} and the gain.

Ex 9.6 drop below the CM

$$V_{th} - V_{ov}$$

folded pair currents (KCL at the fold node)

$$I_{SS2} = I_{SS1} + \frac{I_{SS}}{2}$$

look up (PMOS cascode)

$$R_{up} = g_{m5} r_{O5} r_{O7}$$

look down (two r_o on the fold node)

$$R_{down} = g_{m3} r_{O3} (r_{O1} \parallel r_{O9})$$

transconductance by current divider

$$G_m \approx g_{m1}$$

gain

$$A_v \approx g_{m1} (R_{up} \parallel R_{down})$$

Watch out

- Keep the folding current $\geq I_{SS}$ or a cascode starves in a big step.
- Your page numbers M9, M10 bottom / M7, M8 top; Razavi and Tutorial 2 Q3 swap them. Same formulas.
- Folding costs power, a little gain, an extra pole; it buys swing and a CM range reaching a rail.

Asked in: Tutorial 2 Q3, Problem Set 1 P6–P7, Tutorial 6 Q3.

Lec 6 · Folded-cascode CM ranges, rail-to-rail, folded buffer, low-voltage cascode load

PMOS input: column M11 $\rightarrow$ P $\rightarrow$ M1 $\rightarrow$ X $\rightarrow$ M9, X held at V_{ov9} , link $P = V_{in} + |V_{GS1}|$. Input up: M11 squeezed. Input down: M1 squeezed — and the input can go **below ground**.

PMOS-input folded cascode: the input CM range ($V_{DD} = 1.8\text{ V}$)

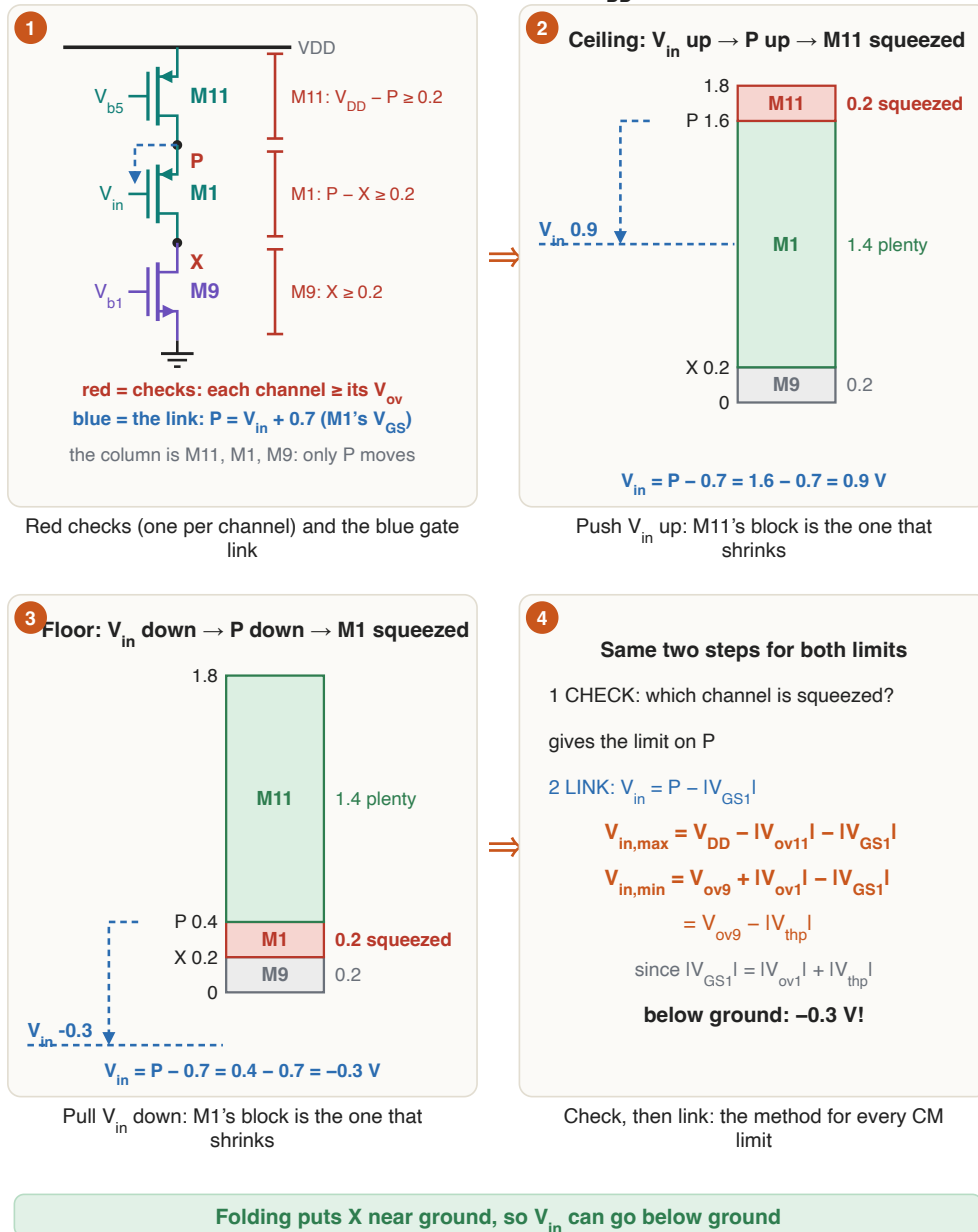
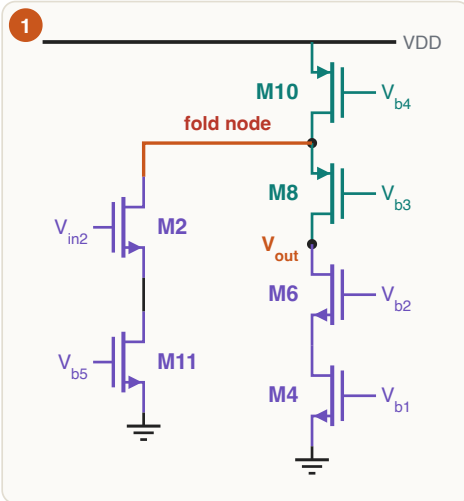


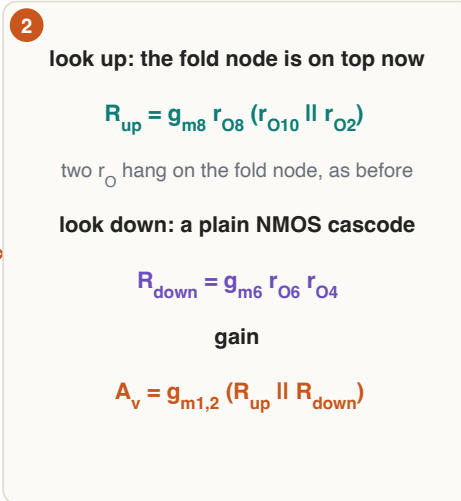
Fig 21. PMOS-input folded cascode CM range: red checks, the blue gate link, and the two towers (ceiling and floor).

NMOS input: everything flips. The fold node hangs from the top, link $S = V_{in} - V_{GS1}$; the ceiling goes **above** V_{DD} .

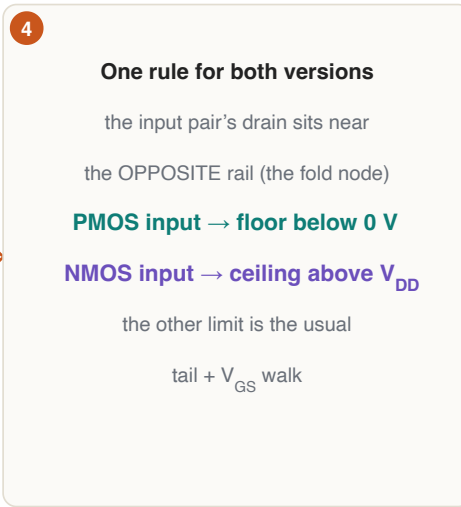
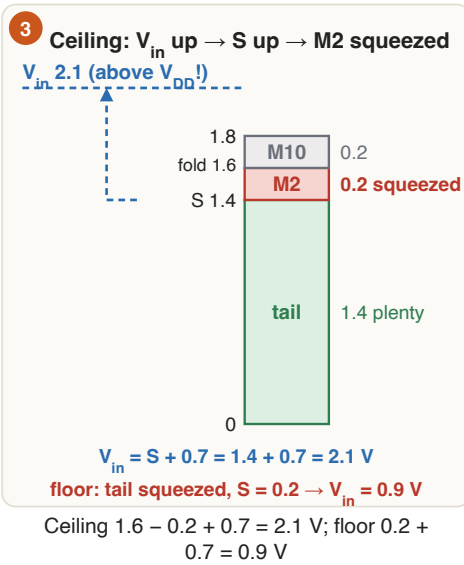
NMOS-input folded cascode: the mirror image



NMOS-input folded cascode, right half (names as on your page)



Gain by the same two looks



Which rail each version can pass

Swap NMOS $\leftrightarrow$ PMOS and every limit flips to the other rail

Fig 22. NMOS-input folded cascode: the circuit, its gain by two looks, and its CM ceiling above V_{DD} .

PMOS-input ceiling

$$V_{in,CM,max} = V_{DD} - |V_{ov11}| - |V_{GS1}|$$

PMOS-input floor (page's two forms)

$$V_{in,CM,min} = V_{ov9} + |V_{ov1}| - |V_{GS1}| = V_{ov9} - |V_{thp}|$$

NMOS-input gain

$$A_v = g_{m1,2} [g_{m8} r_{O8} (r_{O10} \parallel r_{O2}) \parallel g_{m6} r_{O6} r_{O4}]$$

NMOS-input CM range

$$V_{ov11} + V_{GS1} \leq V_{in,CM} \leq V_{DD} - |V_{ov10}| + V_{th1}$$

folded buffer: both checks are floors

$$V_{out} \geq \max(V_{b2} - V_{th4}, V_{b2} - V_{GS4} - |V_{th2}|)$$

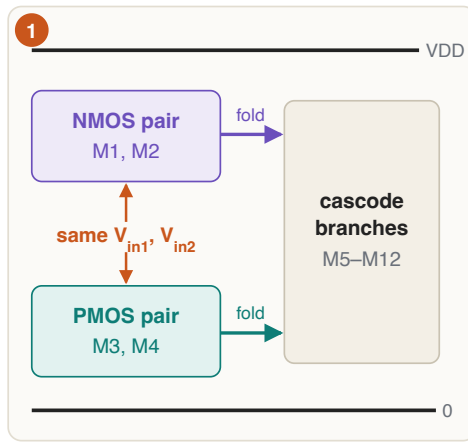
low-voltage cascode load: M5 saturated

$$V_{b1} \geq V_{DD} - |V_{GS7}| - |V_{th5}|$$

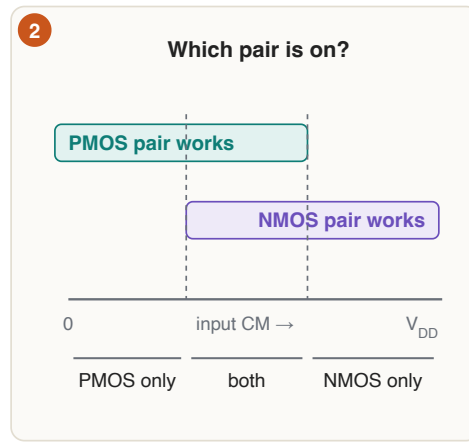
low-voltage cascode load: M7 saturated

$$V_{P,max} = V_{DD} - |V_{ov7}| = V_{DD} - |V_{GS7}| + |V_{th7}|$$

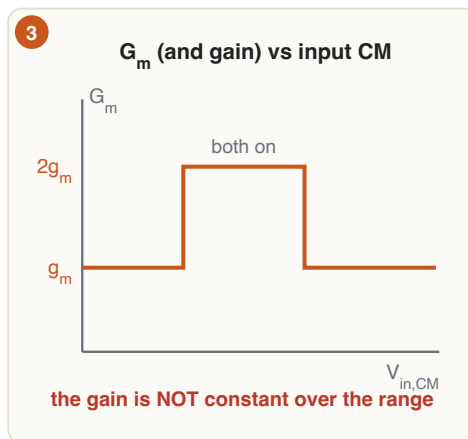
Rail-to-rail input = an NMOS pair + a PMOS pair



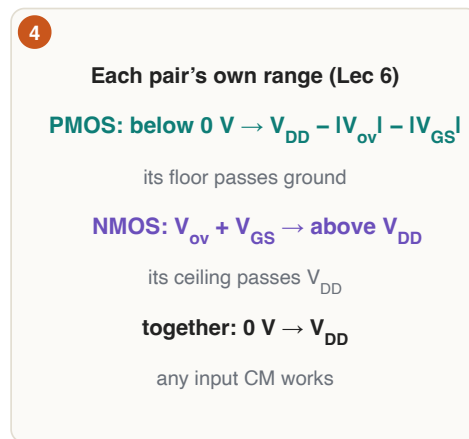
Both pairs fold into the same cascodes



Near 0 V the PMOS pair, near V_{DD} the NMOS pair



Where both work, G_m doubles

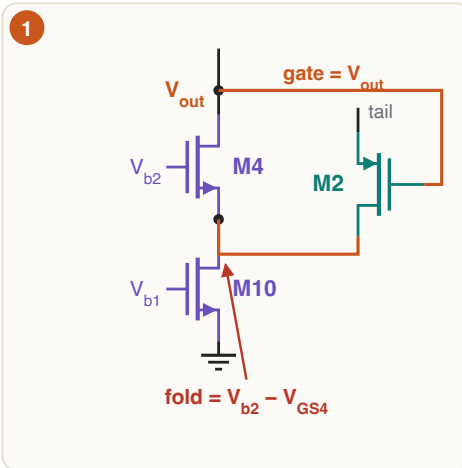


Why the two ranges cover everything

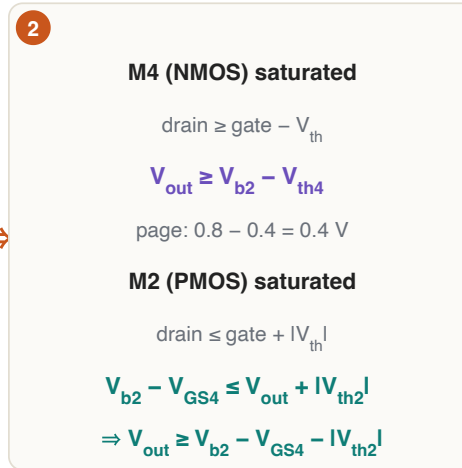
Each pair covers the rail the other cannot reach

Fig 23. Rail-to-rail input: an NMOS and a PMOS pair folded into the same cascodes; G_m doubles in the middle.

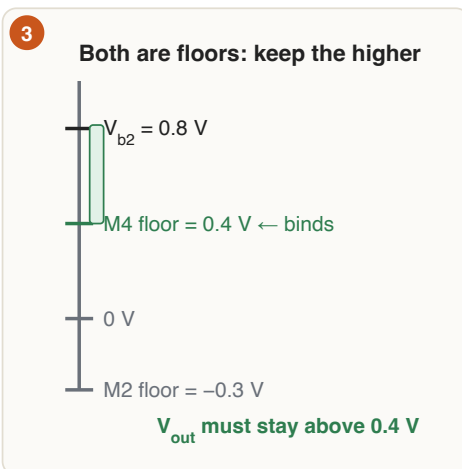
Folded cascode as a unity-gain buffer



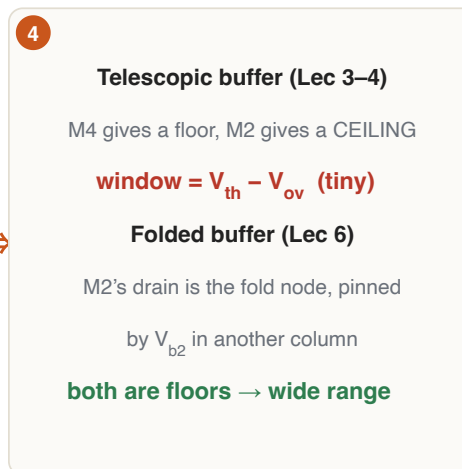
Output tied to M2's gate (the “-” input)



Write each device's fence with V_{out} as M2's gate



Put both floors on one ladder



Why this is far better than the telescopic

Two fences, both floors: the higher one is the real limit

Fig 24. Folded cascode as a buffer: two checks, both floors, the higher one binds.

The low-voltage cascode load (last circuit on your Lec 6 page)

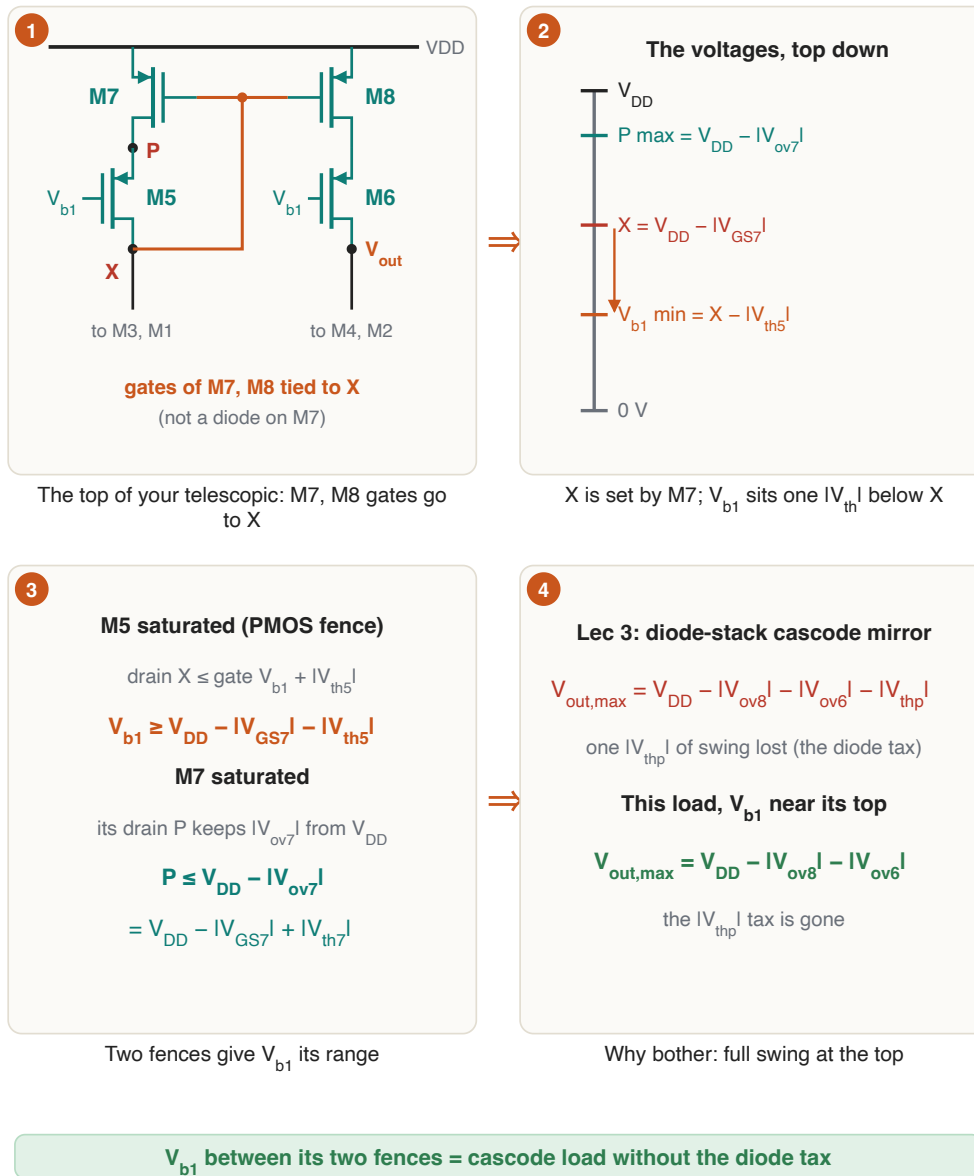


Fig 25. The low-voltage cascode load: M7, M8 gates tied to X; V_{b1} between two fences; no diode tax.

Watch out

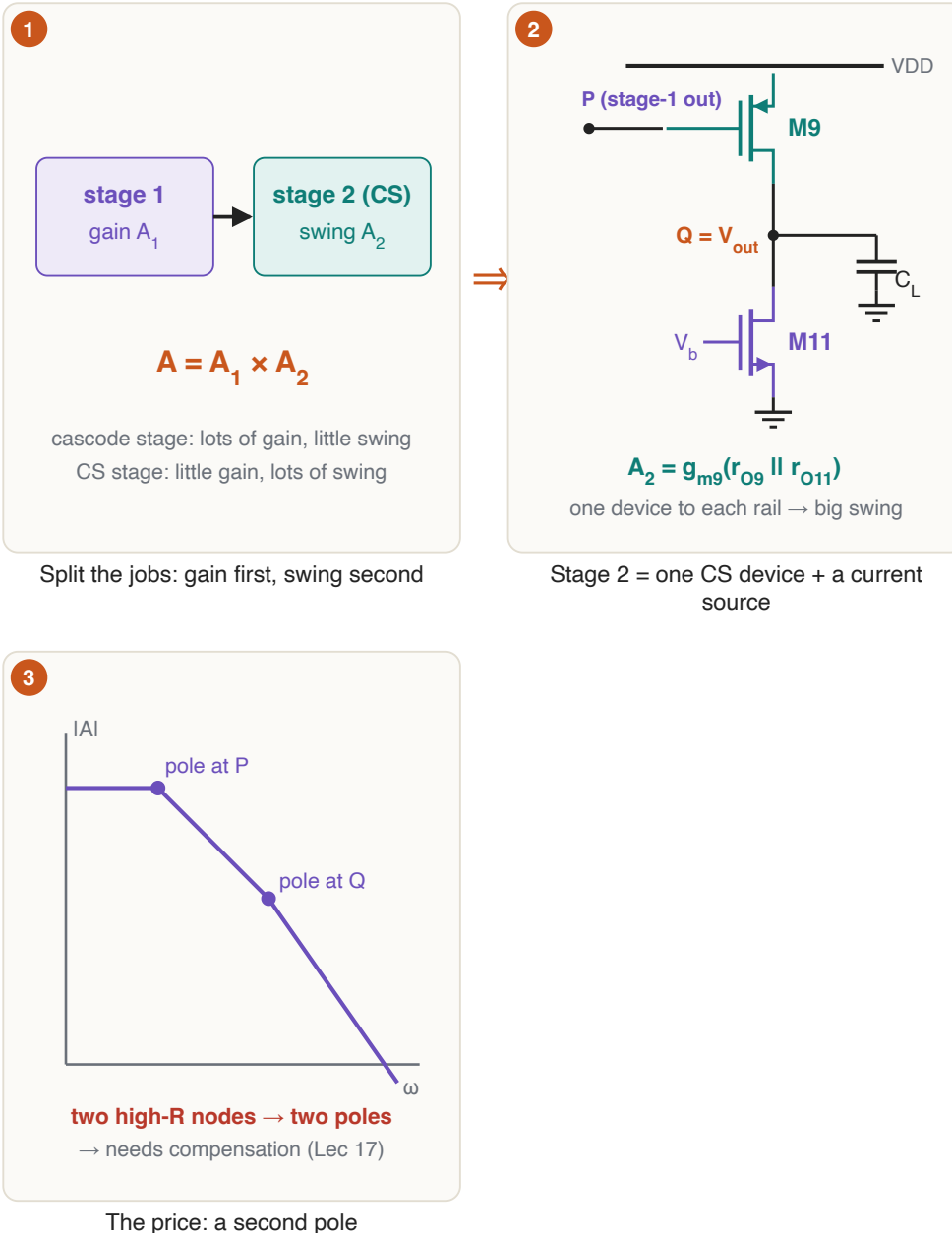
- Page numbers: ceiling $1.8 - 0.2 - 0.7 = 0.9\ V$; floor $0.2 - 0.5 = -0.3\ V$; folded buffer floor $0.8 - 0.4 = 0.4\ V$ binds.
- Rail-to-rail: G_m roughly doubles where both pairs are on.
- Last circuit: M7, M8 gates go to X (not a diode). Your page's $V_{P,max}$ expansion should read $+|V_{th7}|$.
- Why V_{ov} for M11 and V_{GS} for M1: M11's step is a check, M1's step is the link to the gate.

Asked in: Problem Set 1 P8 (folded CM range, minimum negative); Quiz 1 2023-24 Q2.

Lec 7 · Two-stage op amps and the idea of gain boosting

Split the jobs: stage 1 (cascode or pair) for **gain**, stage 2 (a CS stage: one device and one current source) for **swing**. Gains multiply. Price: two high-resistance nodes $\rightarrow$ two poles $\rightarrow$ compensation (Lec 17).

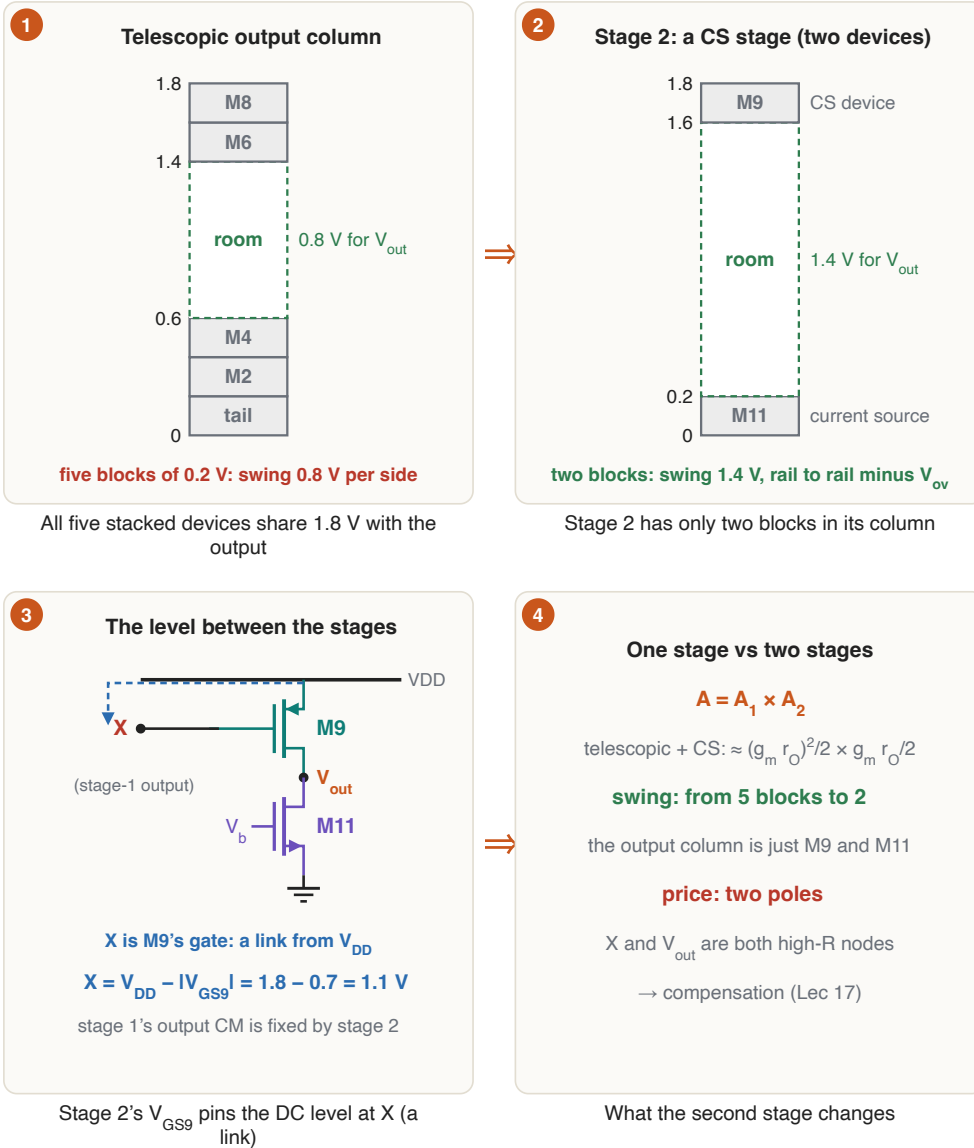
Two-stage op amp, simplified



Gains multiply; every high-resistance node adds a pole

Fig 26. Two-stage op amp: gain first, swing second — and the price, a second pole.

Why two stages: towers at $V_{DD} = 1.8\text{ V}$ ($V_{ov} = 0.2\text{ V}$ each)



Stage 1 gives the gain, stage 2 gives the swing

Fig 27. Why two stages: the telescopic's five blocks vs the CS stage's two, and the link that sets the level between the stages.

simple two-stage

$$A_1 = g_{m1,2}(r_{O1,2} \parallel r_{O3,4}), \quad A_2 = g_{m5,6}(r_{O5,6} \parallel r_{O7,8}), \quad A = A_1 A_2$$

telescopic first stage

$$A_1 = g_{m1} [g_{m5} r_{O5} r_{O7} \parallel g_{m3} r_{O3} r_{O1}]$$

PMOS CS second stage

$$A_2 = g_{m9}(r_{O9} \parallel r_{O11})$$

level between the stages (a link)

$$X = V_{DD} - |V_{GS9}|$$

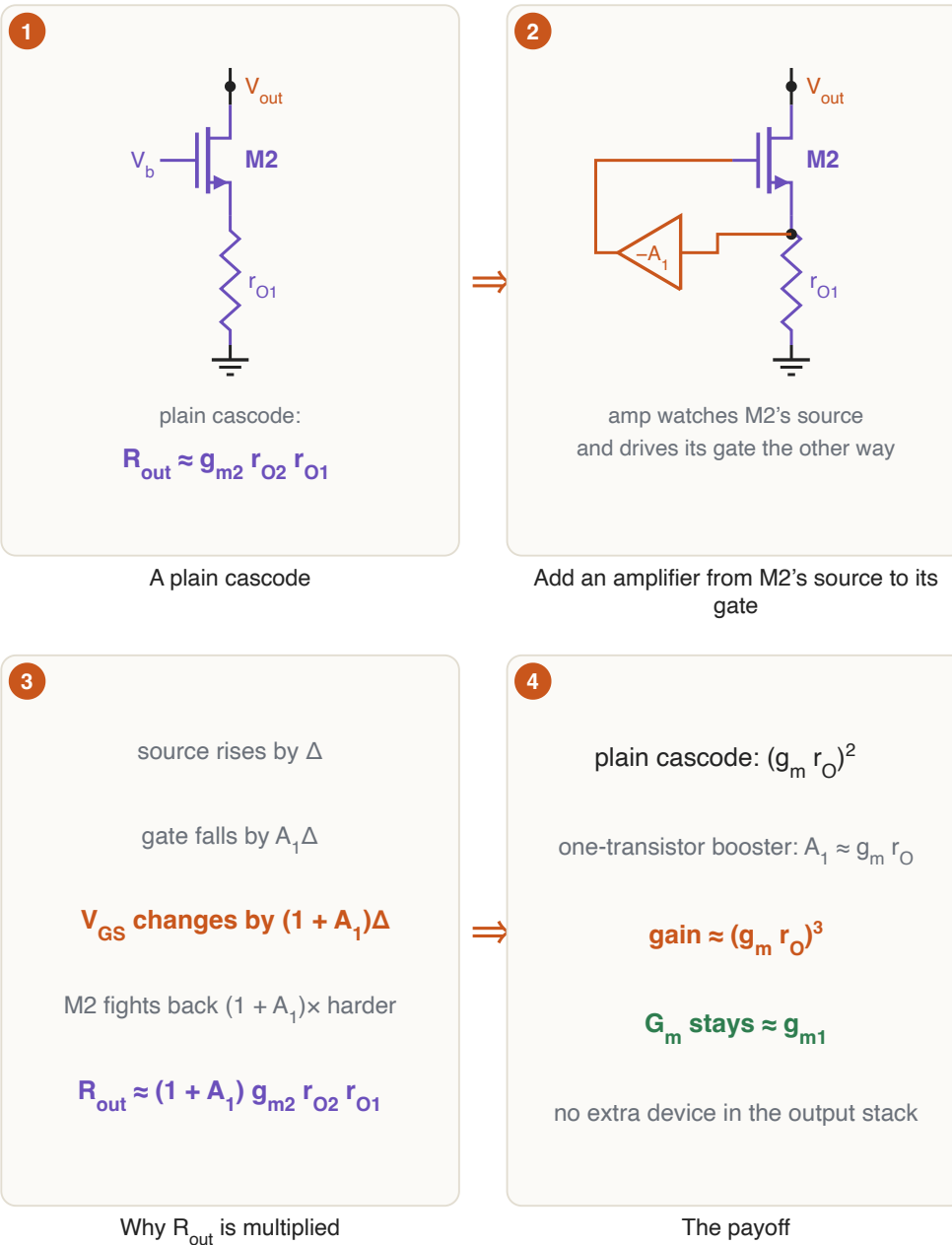
every gain

$$A_v = G_m \times R_{out}$$

boosted output resistance (test source)

$$R_{out} = R_S + r_O + (1 + A_1)g_m R_S r_O$$

Gain boosting, simplified



Boosting multiplies R_{out} by $(1 + A_1)$ without stacking another device

Fig 28. Gain boosting simplified: a cascode $\Rightarrow$ add an amplifier from source to gate $\Rightarrow$ why R_{out} is multiplied by $(1 + A_1) \Rightarrow (g_m r_O)^3$.

Watch out

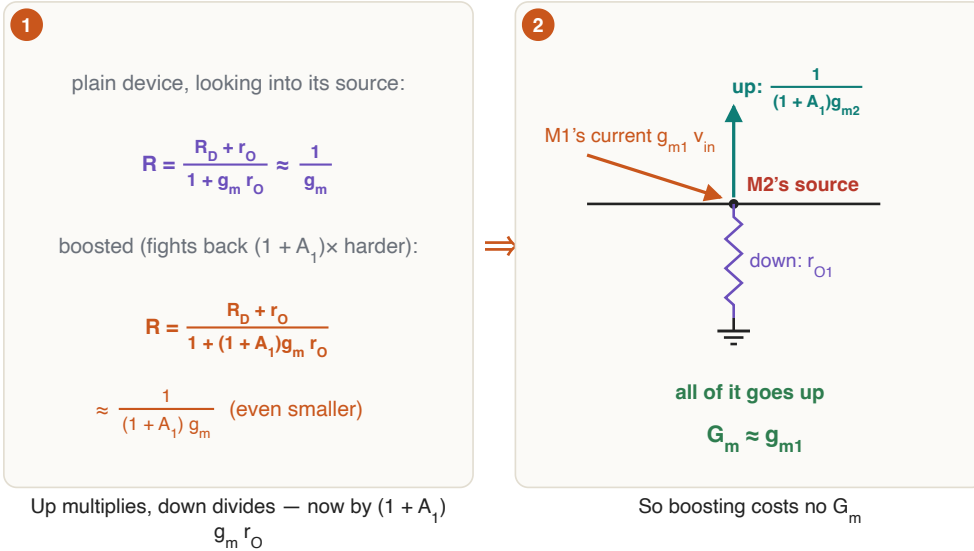
- Gain first, swing second: stage 1's output only moves by V_{out}/A_2 .
- An amplifier in **front** of the gate does not make the stage better ("no improvement"); the amplifier must fight the **output** resistance.
- Test-source method: input off, apply V_X at the output, $R_{out} = V_X/I_X$.

Asked in: Tutorial 3 Q3 (Razavi Fig 9.24): CM level at X, Y, sizes for the swing, overall gain; Tutorial 4 Q1–Q2.

Lec 8 · Gain boosting: G_m , R_{out} , $(g_m \cdot r_O)^3$ and three boosters

With the amplifier sensing the source, A_1 cancels in G_m (it becomes $\approx 1/R_S$), so boosting is purely an R_{out} multiplier. The booster senses M2's source X and drives M2's gate.

Looking into a boosted source



A boosted source is even stiffer: the input current all reaches the output

Fig 29. Looking into a boosted source: $(1 + A_1)$ times stiffer, so the input current all goes up ($G_m \approx g_{m1}$).

What gain boosting changes, and what it does not

	plain cascode	gain-boosted cascode
G_m	$\approx g_{m1}$	$\approx g_{m1}$ (same)
R_{out}	$g_{m2} r_{O2} r_{O1}$	$(1 + A_1) g_{m2} r_{O2} r_{O1}$
gain	$(g_m r_O)^2$	$(g_m r_O)^3$ when $A_1 = g_m r_O$
output column	M2 on M1	M2 on M1, nothing stacked (same)
lowest V_{out}	$V_{ov1} + V_{ov2}$	CS booster: $V_{GS3} + V_{ov2}$; folded: no loss
extra cost	—	booster current and one more node
booster wiring	—	senses M2's source X, drives M2's gate

Green = unchanged, red = what boosting changes

Same G_m and $R_{out} \times (1 + A_1)$, with no extra device in the output column

Fig 30. Plain vs gain-boosted cascode: what boosting changes and what it leaves alone.

G_m with sensing (no help)

$$\frac{I_{out}}{V_{in}} \approx \frac{A_1 g_m}{R_S + (1 + A_1) g_m r_O R_S} \approx \frac{1}{R_S}$$

boosted R_{out} , $R_S = r_{O1}$

$$R_{out} \approx (1 + A_1) g_{m2} r_{O2} r_{O1}$$

into a boosted source

$$R_{in,source} = \frac{R_D + r_O}{1 + (1 + A_1) g_m r_O}$$

boosted cascode gain

$$A_v \approx g_{m1} (1 + A_1) g_{m2} r_{O2} r_{O1} \approx (g_m r_O)^3$$

CS booster

$$A_1 = g_{m3} r_{O3}, \quad V_{out,min} = V_{GS3} + V_{ov2}, \quad V_{G2} = V_{GS3} + V_{GS2}$$

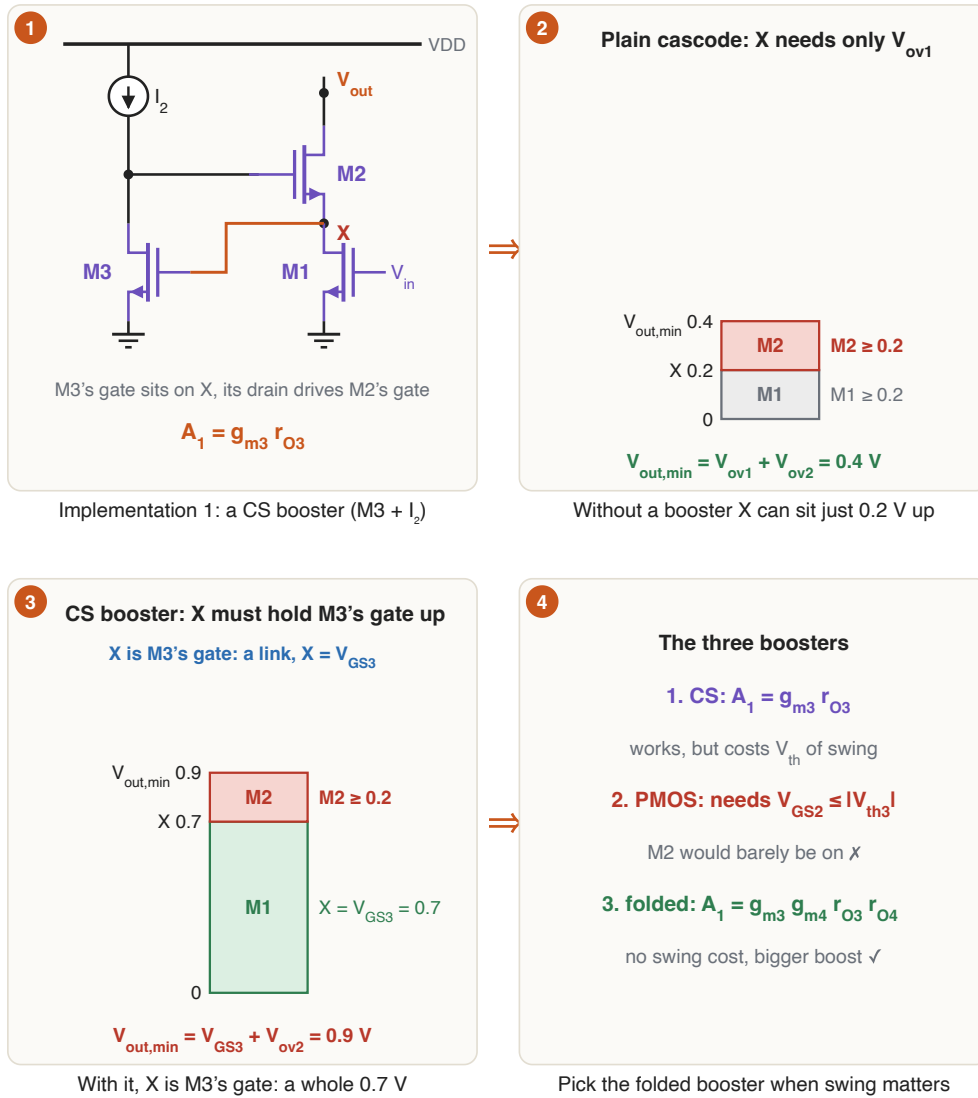
PMOS booster (fails)

$$V_{GS2} \leq |V_{th3}|$$

folded booster

$$A_1 = g_{m3} g_{m4} r_{O4} r_{O3}$$

Building the booster, and what it costs



The CS booster's M1 block is stuck at V_{GS3} : half a volt of swing gone

Fig 31. The CS booster as towers: X is M3's gate, so it sits a whole V_{GS3} up and the output loses one V_{th} of swing.

Watch out

- Cascode $(g_m r_O)^2$, boosted cascode $(g_m r_O)^3$.
- CS booster: X is M3's gate (a link), so $X = V_{GS3}$ instead of V_{ov1} : one V_{th} of swing lost.
- $G_m \approx g_{m1}$ still: the boosted source is only $\approx 1 / ((1 + A_1)g_{m2})$.

Asked in: Tutorial 4 Q1(a)–(c) (bias of M2, M3; Rout; swing), Q2(e), Q3 (folded booster).

Formula index: every formula above, in lecture order

Lec 1

feedback factor (tap across the bottom resistor)

$$\beta = \frac{R_2}{R_1 + R_2}$$

closed-loop gain

$$A_{closed} = \frac{A}{1 + \beta A}$$

ideal gain

$$A_{ideal} = \frac{1}{\beta} = 1 + \frac{R_1}{R_2}$$

gain error

$$\varepsilon = \frac{1}{1 + \beta A} \approx \frac{1}{\beta A}$$

minimum open-loop gain

$$A_{min} = \frac{A_{closed}}{\varepsilon}$$

Lec 2

gain-bandwidth product

$$\omega_u = A_0 \omega_0 = \text{GBW}$$

fully differential pair, PMOS current-source loads

$$A_v = g_{m1,2}(r_{O1,2} \parallel r_{O3,4})$$

its input CM range

$$V_{ISS} + V_{GS1} \leq V_{in,CM} \leq V_{DD} - |V_{ov3}| + V_{thn}$$

its differential swing (x2: outputs move oppositely)

$$2(V_{DD} - |V_{ov3}| - V_{ov1} - V_{ISS})$$

output pole

$$\omega_p = \frac{1}{(r_{O2} \parallel r_{O4})C_L}$$

5-T OTA gain

$$A_v = g_{m2}(r_{O2} \parallel r_{O4})$$

CM ceiling (diode node, then one V_{th} up)

$$V_{in,CM,max} = V_{DD} - |V_{GS3}| + V_{th1}$$

CM floor (tail, then a whole V_{GS})

$$V_{in,CM,min} = V_{ISS} + V_{GS1}$$

single-ended swing (no x2)

$$V_{DD} - |V_{ov4}| - V_{ov2} - V_{ISS}$$

Lec 3

buffer gain

$$A_{closed} = \frac{A}{1 + A} \approx 1$$

closed-loop output resistance

$$R_{out,closed} = \frac{R_{out,open}}{1 + \beta A} \approx \frac{1}{g_{m2}}$$

buffer bandwidth = the GBW

$$\omega = \frac{g_{m2}}{C_L}$$

telescopic gain

$$A = g_{m1,2} [g_{m4}r_{O4}r_{O2} \parallel g_{m6}r_{O6}r_{O8}] \approx \frac{(g_m r_O)^2}{2}$$

differential swing (five blocks per column)

$$2[V_{DD} - (|V_{ov8}| + |V_{ov6}| + V_{ov4} + V_{ov2} + V_{ISS})]$$

mirror-loaded version (diode tax)

$$V_{DD} - (|V_{ov8}| + |V_{ov6}| + |V_{thp}| + V_{ov4} + V_{ov2} + V_{ISS})$$

buffer window (V_{out} on M2's gate)

$$V_{b1} - V_{th4} \leq V_{out} \leq V_{b1} - V_{GS4} + V_{th2}$$

window width

$$V_{th} - V_{ov4}$$

Lec 4

currents

$$I_{total} = \frac{P}{V_{DD}}$$

swing budget (per column)

$$|V_{ov8}| + |V_{ov6}| + V_{ov4} + V_{ov2} + V_{ov9} = V_{DD} -$$

swing per side

square law

$$\frac{W}{L} = \frac{2I_D}{\mu C_{ox} V_{ov}^2}$$

small-signal

$$g_m = \frac{2I_D}{V_{ov}}, \quad r_O = \frac{1}{\lambda I_D}$$

gain check

$$A_v = g_{m1,2}(R_{up} \parallel R_{down})$$

fix without touching swing

$$g_m r_O \propto \sqrt{\frac{WL}{I_D}}, \quad \lambda \propto \frac{1}{L}$$

Lec 5

Ex 9.6 drop below the CM

$$V_{th} - V_{ov}$$

folded pair currents (KCL at the fold node)

$$I_{SS2} = I_{SS1} + \frac{I_{SS}}{2}$$

look up (PMOS cascode)

$$R_{up} = g_{m5} r_{O5} r_{O7}$$

look down (two r_{O1} on the fold node)

$$R_{down} = g_{m3} r_{O3} (r_{O1} \parallel r_{O9})$$

transconductance by current divider

$$G_m \approx g_{m1}$$

gain

$$A_v \approx g_{m1} (R_{up} \parallel R_{down})$$

Lec 6

PMOS-input ceiling

$$V_{in,CM,max} = V_{DD} - |V_{ov11}| - |V_{GS1}|$$

PMOS-input floor (page's two forms)

$$V_{in,CM,min} = V_{ov9} + |V_{ov1}| - |V_{GS1}| = V_{ov9} - |V_{thp}|$$

NMOS-input gain

$$A_v = g_{m1,2} [g_{m8} r_{O8} (r_{O10} \parallel r_{O2}) \parallel g_{m6} r_{O6} r_{O4}]$$

NMOS-input CM range

$$V_{ov11} + V_{GS1} \leq V_{in,CM} \leq V_{DD} - |V_{ov10}| + V_{th1}$$

folded buffer: both checks are floors

$$V_{out} \geq \max(V_{b2} - V_{th4}, V_{b2} - V_{GS4} - |V_{th2}|)$$

low-voltage cascode load: M5 saturated

$$V_{b1} \geq V_{DD} - |V_{GS7}| - |V_{th5}|$$

low-voltage cascode load: M7 saturated

$$V_{P,max} = V_{DD} - |V_{ov7}| = V_{DD} - |V_{GS7}| + |V_{th7}|$$

Lec 7

simple two-stage

$$A_1 = g_{m1,2} (r_{O1,2} \parallel r_{O3,4}), \quad A_2 = g_{m5,6} (r_{O5,6} \parallel r_{O7,8}), \quad A = A_1 A_2$$

telescopic first stage

$$A_1 = g_{m1} [g_{m5} r_{O5} r_{O7} \parallel g_{m3} r_{O3} r_{O1}]$$

PMOS CS second stage

$$A_2 = g_{m9} (r_{O9} \parallel r_{O11})$$

level between the stages (a link)

$$X = V_{DD} - |V_{GS9}|$$

every gain

$$A_v = G_m \times R_{out}$$

boosted output resistance (test source)

$$R_{out} = R_S + r_O + (1 + A_1) g_m R_S r_O$$

Lec 8

G_m with sensing (no help)

$$\frac{I_{out}}{V_{in}} \approx \frac{A_1 g_m}{R_S + (1 + A_1) g_m r_O R_S} \approx \frac{1}{R_S}$$

boosted R_{out}, R_S = r_{O1}

$$R_{out} \approx (1 + A_1) g_{m2} r_{O2} r_{O1}$$

into a boosted source

$$R_{in,source} = \frac{R_D + r_O}{1 + (1 + A_1) g_m r_O}$$

boosted cascode gain

$$A_v \approx g_{m1} (1 + A_1) g_{m2} r_{O2} r_{O1} \approx (g_m r_O)^3$$

CS booster

$$A_1 = g_{m3} r_{O3}, \quad V_{out,min} = V_{GS3} + V_{ov2}, \quad V_{G2} = V_{GS3} + V_{GS2}$$

PMOS booster (fails)

$$V_{GS2} \leq |V_{th3}|$$

folded booster

$$A_1 = g_{m3} g_{m4} r_{O4} r_{O3}$$